ON FRIEDMAN'S PROPERTY

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Abstract. We define forcing orders which add witnesses to the failure of various forms of Friedman's Property. These posets behave similarly to the forcing order adding a nonreflecting stationary set but have the advantage of allowing the construction of master conditions and thus the preservation of various large cardinal properties. We apply these new techniques to separate various instances of variants of Friedman's Problem, both between different instances at one cardinal as well as equal instances at different cardinals and en passant obtain some new results regarding the differences between $<\kappa$- and κ-strategic closure.

In [7], Harvey Friedman proved the important Theorem that any stationary subset of ω_1 contains closed subsets of arbitrarily long, countable, ordertype. In that same paper he introduces the property $F(\kappa)$ for $\kappa \geq \omega_2$, which states that any subset of κ either contains or is disjoint from a closed set of ordertype ω_1. We note that ω_1 clearly is the critical ordinal since the same statement for ordinals $<\omega_1$ is provable in ZFC and the same statement for ordinals $>\omega_1$ is inconsistent (take $A := E^\kappa_\omega$). Later on, the stronger property $F^+(\kappa)$, stating that every stationary subset of E^κ_ω contains a closed subset of ordertype ω_1, was shown by Shelah in [19] (Chapter XI, Theorem 7.1) to be consistent from a Mahlo cardinal and in [5] and [4] to follow from MM and SRP respectively. It was also noticed by Silver (as stated in [7]) that forcing the failure of $F(\kappa)$ is quite easy, as any model of the form $V[\mathrm{Coll}(\omega, \omega_1)]$ will satisfy $\neg F(\kappa)$ for any $\kappa > \omega_1$.

In this work, we will introduce posets which add witnesses to the failure of various variants of F and F^+ in a more gentle manner in order to separate many instances of F and F^+. These posets are modelled after the poset adding a nonreflecting stationary set but have the crucial advantage that they allow the building of master conditions and thus the lifting of ground-model embeddings. This is used in two ways: By forcing with our new posets over the original model of Martin's Maximum obtained in [5], we obtain a model where some form of F or F^+ (and thus also MM) fails, but Martin's Maximum holds for all posets which do not imply the corresponding form of F or F^+. Using this we will obtain a perfect separation between instances of F or F^+ at ω_2 (i.e. we prove implications between instances of F and F^+ and show that all other implications are not provable in ZFC). We will also use our posets to obtain a separation between instances of F or F^+ at different cardinals. Here we can show that, despite their similarities, F and F^+ have different "Meta-properties" thanks to the increased complexity of F^+ compared to F. For example, we will show that $F^+(\delta)$ can hold for the first time at λ^+, where κ is λ-supercompact, while $F(\lambda^+)$ must fail, provided κ is λ-supercompact and $F(\delta)$ fails for all $\delta < \kappa$.

The paper is organized as follows: In the first section, we define the notions that will appear in this paper and give a more detailed introduction into our results. In

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Section 2, we introduce our posets to add witnesses to the failure of F and F^+. In Section 3, we introduce the new maximal versions of MM and prove their relative consistency. In Section 4 and Section 5, we use these axioms to obtain separations between instances of F and F^+ respectively. In Section 6, we show that we can force F and F^+ to fail in a more gentle way while preserving large cardinals and obtain results regarding the compactness of the failure of F and F^+ at large cardinals. We close with some open questions.

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1. Introduction and Preliminaries

We assume the reader has some familiarity with the basic theory of club and stationary sets as well as a solid forcing knowledge. Our notation is standard, we require filters to be upwards closed (so that $q \leq p$ implies that q forces more than p). For any ordinal γ we let E^γ_ω denote the set of all ordinals $\delta \in \gamma$ with $\mathrm{cf}(\delta) = \omega$. A function is said to be *normal* if it is strictly increasing and continuous.

We first define our variants of F and F^+:

Definition 1.1. Let $\kappa \geq \omega_2$ be regular.

(1) Let $\lambda \leq \kappa$ be a cardinal. The property $F(\lambda, \kappa)$ states that whenever $g\colon \kappa \to \lambda$ is regressive there is a normal function $f\colon \omega_1 \to \kappa$ such that $g \circ f$ is constant. $F(\kappa)$ is defined as $F(2, \kappa)$.

(2) Let $(D_i)_{i\in\omega_1}$ be a partition of ω_1 (we allow $D_i = \emptyset$ for some i). The property $F^+((D_i)_{i\in\omega_1}, \kappa)$ states that whenever $(A_i)_{i\in\omega_1}$ is a sequence of stationary subsets of E^κ_ω there is a normal function $f\colon \omega_1 \to \kappa$ such that $f[D_i] \subseteq A_i$ for all $i \in \omega_1$. $F^+(\kappa)$ is defined as $F((D_i)_{i\in\omega_1}, \kappa)$, where $D_0 = \omega_1$.

Regarding the consistencies of these properties, it was first shown by Foreman, Magidor and Shelah in their famous paper on Martin's Maximum [5] that MM implies $F^+(\kappa)$ for every regular $\kappa \geq \omega_2$. This was later improved by Feng and Jech [4] who showed that the *strong reflection principle* SRP, a consequence of MM, implies $F^+((D_i)_{i\in\omega_1}, \kappa)$ whenever $\kappa \geq \omega_2$ is regular and $(D_i)_{i\in\omega_1}$ is a maximal partition of ω_1 into stationary sets (i.e. every D_i is stationary and for every stationary $S \subseteq \omega_1$ the set $S \cap D_i$ is stationary for some $i \in \omega_1$). We will later (see Theorem 5.2) improve their result by showing that SRP even implies $F^+((D_i)_{i\in\omega_1}, \kappa)$ for every regular $\kappa \geq \omega_2$ and *every* partition $(D_i)_{i\in\omega_1}$, answering a question of Fuchs (see [8, Section 5]).

For $\kappa = \omega_2$, the principles have much lower consistency strength than the supercompact cardinal used to obtain Martin's Maximum: Shelah showed (see [19, Chapter XI, Theorem 7.1 and Theorem 7.2]) that one can force $F^+((D_i)_{i\in\omega_1}, \omega_2)$ for any partition $(D_i)_{i\in\omega_1}$ from a weakly compact cardinal and $F^+((D_i)_{i\in\omega_1}, \omega_2)$ for the partition with $D_0 = \omega_1$ from a Mahlo cardinal. We will give an easier proof of the second result in this paper.

These properties are of course related to the more well-known principle of *stationary reflection*:

Definition 1.2. Let κ be a regular cardinal and $S \subseteq \kappa$ stationary. $\mathrm{SR}(S)$ states that whenever $T \subseteq S$ is stationary, there is $\gamma < \kappa$ of uncountable cofinality such that $T \cap \gamma$ is stationary in γ.

Remark 1.3. The previous three principles are connected as follows:

(1) $F^+(\kappa)$ implies that every stationary subset of E^κ_ω reflects to an ordinal γ with cofinality ω_1. In particular, $F^+(\omega_2)$ implies that the *square principle* $\square_{\omega_1}$ fails (see e.g. [15, Lemma 23.6]). This, in turn, implies that ω_2 is Mahlo in L by a classical result of Jensen (see [16]).

On the other hand, if $(D_i)_{i\in\omega_1}$ is a nontrivial partition of ω_1 (i.e. at least two D_i are stationary in ω_1), then $F^+((D_i)_{i\in\omega_1},\omega_2)$ implies that any two stationary subsets of $E^{\omega_2}_\omega$ reflect to the same ordinal γ. This, in turn, implies that ω_2 is weakly compact in L by a classical result of Magidor (see [18]).

So the consistency strength of $F^+(\omega_2)$ is at least a Mahlo cardinal and that of $F^+((D_i)_{i\in\omega_1},\omega_2)$ (if $(D_i)_{i\in\omega_1}$ is nontrivial) is at least a weakly compact cardinal. By results of Shelah (see [19, Chapter XI, Theorem 7.1 and Theorem 7.2]), this is the exact consistency strength.

(2) Clearly, $F^+(\kappa)$ implies $F(\kappa,\kappa)$, since, by Fodor's Lemma, every regressive function on E^κ_ω in particular is constant on a stationary subset of E^κ_ω which then contains a closed copy of ω_1.

(3) By Silver's result, it is consistent that there are e.g. weakly compact cardinals and $F(\delta)$ fails for all δ. In particular, $\mathrm{SR}(E^\kappa_\omega)$ does not imply $F(\kappa)$. We will also see in Remark 2.8 that $F(\kappa)$ does not imply $\mathrm{SR}(E^\kappa_\omega)$.

For $F(\lambda,\kappa)$, it is clear that $F(\lambda,\kappa)$ implies $F(\lambda',\kappa)$ whenever $\lambda'<\lambda$ (as any function from κ to λ' is also a function from $\kappa\to\lambda$). Additionally, $F(\lambda,\kappa)$ implies $F(\lambda,\kappa')$ whenever $\omega_2\leq\kappa'<\kappa$ (by simply restricting the function witnessing $F(\lambda,\kappa)$). We will show that the first implication cannot be reversed:

Theorem A. *Assume κ is supercompact and $\lambda\leq\aleph_1$ is a cardinal (which may be finite). There is a forcing extension in which $\kappa=\omega_2$, $F(\lambda,\kappa)$ holds and $F(\lambda',\kappa)$ fails for any $\lambda'>\lambda$.*

So it is e.g. relatively consistent that any partition of ω_2 into 42 pieces has at least one part that includes a closed copy of ω_1 while there is a partition of ω_2 into 43 sets such that no part includes a closed copy of ω_1.

For the principle F^+, the implications are more intricate. Given partitions $\overline{D}=(D_i)_{i\in\omega_1}$ and $\overline{E}=(E_i)_{i\in\omega_1}$, we say that $\overline{D}\leq\overline{E}$ if $\overline{D}$ refines $\overline{E}$ and $\overline{D}\leq^*\overline{E}$ if $\overline{D}$ refines $\overline{E}$ modulo a nonstationary set. The latter partial order, $\leq^*$, exactly characterizes when there is an implication between $F^+((D_i)_{i\in\omega_1})$ and $F^+((E_i)_{i\in\omega_1})$:

Theorem B. *Let $\overline{D}$ be a partition of ω_1.*

(1) *If $\overline{E}$ is a partition of ω_1 with $\overline{E}\leq^*\overline{D}$ and $\kappa\geq\omega_2$ is regular, $F^+(\overline{E},\kappa)$ implies $F^+(\overline{D},\kappa)$.*

(2) *Assume V is the model satisfying* MM *obtained in [5]. Then there is a forcing extension in which $F^+(\overline{D},\omega_2)$ fails and $F^+(\overline{E},\omega_2)$ holds for all partitions $\overline{E}$ with $\overline{E}\not\leq^*\overline{D}$ and all regular $\Theta\geq\kappa$.*

In fact, we will even obtain that the canonical fragment of SRP associated with $F^+(\overline{E},\omega_2)$ holds, answering a question of Fuchs from [8].

Definition 1.4. MM states that whenever $\mathbb{P}$ is a forcing notion which preserves stationary subsets of ω_1 and $(D_i)_{i\in\omega_1}$ is a sequence of open dense subsets of $\mathbb{P}$ there is a filter $G\subseteq\mathbb{P}$ such that $G\cap D_i\neq\emptyset$ for any $i\in\omega_1$.

A very important consequence of MM which captures many of its implications is the principle SRP, introduced by Todorčević in a handwritten note in 1987. Later on, an equivalent, but simpler, characterization was found by Feng and Jech in [4]. Recently, in [8], Gunter Fuchs introduced the parametrized variants.

Definition 1.5. Let Θ be a regular cardinal. $S \subseteq [H(\Theta)]^{<\omega_1}$ is *projective stationary* if for any stationary $A \subseteq \omega_1$, the set

$$\{M \in S \mid M \cap \omega_1 \in A\}$$

is stationary in $[H(\Theta)]^{<\omega_1}$.

The *strong reflection principle* SRP states that whenever S is projective stationary, there is an increasing and continuous $\in$-chain $(N_i)_{i\in\omega_1}$ of members of S.

If $\mathcal{S}$ is a collection of projective stationary subsets of $[H(\Theta)]^{<\omega_1}$, SRP($\mathcal{S}$) states that whenever $S \in \mathcal{S}$ there is an increasing and continuous $\in$-chain $(N_i)_{i\in\omega_1}$ of members of S.

Given a partition $(D_i)_{i\in\omega_1}$ of ω_1 and a sequence $(A_i)_{i\in\omega_1}$ of stationary subsets of E^κ_ω there is (for $\Theta \geq \kappa$) a canonical subset of $[H(\Theta)]^{<\omega_1}$ associated with the pair $((D_i)_{i\in\omega_1}, (A_i)_{i\in\omega_1})$: We let

$$S((D_i)_{i\in\omega_1}, (A_i)_{i\in\omega_1}, \Theta) := \\ \{M \in [H(\Theta)]^{<\omega_1} \mid \forall i \in \omega_1 (M \cap \omega_1 \in D_i \to \sup(M \cap \kappa) \in A_i)\}$$

By an argument of Fuchs from [8], if there is an increasing and continuous $\in$-chain $(N_i)_{i\in\omega_1}$ of members of $S((D_i)_{i\in\omega_1}, (A_i)_{i\in\omega_1}, \Theta)$ for some Θ then there is a normal function $f\colon \omega_1 \to \kappa$ such that $f[D_i] \subseteq A_i$ (by simply letting $f(N_i \cap \omega_1) := \sup(N_i \cap \kappa)$ and "filling in the gaps", see the proof of Theorem 5.2). We will later show, answering a question of Fuchs from the same paper, that $S((D_i)_{i\in\omega_1}, (A_i)_{i\in\omega_1}, \Theta)$ is always projective stationary, which means that SRP implies $F^+((D_i)_{i\in\omega_1}, \kappa)$ for any partition $(D_i)_{i\in\omega_1}$ of ω_1 and any regular cardinal $\kappa \geq \omega_2$.

We will also define $\mathcal{S}((D_i)_{i\in\omega_1}, \kappa, \Theta)$ as the collection of $S((D_i)_{i\in\omega_1}, (A_i)_{i\in\omega_1}, \Theta) \cap C$ for any sequence $(A_i)_{i\in\omega_1}$ of stationary subsets of E^κ_ω and any club $C \subseteq [H(\Theta)]^{<\omega_1}$. In section 5 we answer a question of Fuchs [8] in a strong way by showing that whenever $\overline{D}$ is a partition of ω_1 in the original model of MM obtained in [5], it is consistent that $F^+(\overline{D}, \omega_2)$ fails but SRP($\mathcal{S}(\overline{E}, \kappa, \Theta)$) holds for all regular κ, $\Theta \geq \kappa$ and partitions $\overline{E}$ with $\overline{D} \not\leq^* \overline{E}$.

In the last section we will concern ourselves with the effect large cardinal properties can have on the pattern of F and F^+. We will show that there is a natural separation between F and F^+ caused by the different complexity ($\neg F(\kappa)$ is a Π^1_1-statement while $\neg F^+$ is more complex due to the requirement of stationarity). This is also reflected in the different closure properties of the corresponding forcings (a similar situation occurs in the separation between the existence of a square sequence and a nonreflecting stationary set, see Remark 6.7 in Cummings' chapter in the Handbook of Set Theory [1]):

Definition 1.6. Let $\mathbb{P}$ be a poset and γ an ordinal. The *completeness game* $G(\mathbb{P}, \gamma)$ on $\mathbb{P}$ of length γ is played by COM and INC as follows: COM plays at even ordinals (including limits) and INC plays at odd ordinals. The first move by COM is $1_\mathbb{P}$. Afterwards, given a sequence $(p_\alpha)_{\alpha<\delta}$ for $\delta < \gamma$, the player whose turn it is needs to play a condition p_δ such that $p_\delta \leq p_\alpha$ for any $\alpha < \delta$. COM wins if they have a legal move at every ordinal below (but not including) γ, otherwise INC wins.

We say that a forcing notion is γ-strategically closed if COM has a winning strategy in $G(\mathbb{P}, \gamma)$. It is $<\gamma$-strategically closed if it is δ-strategically closed for all $\delta < \gamma$.

The completeness game first occurs in work of Jech [14]. Since we are later interested in the distinction between $<\kappa$-strategic closure and κ-strategic closure, it is also interesting to note the following results by Velleman [21] and Ishiu-Yoshinobu [12]: Velleman showed that, if Jensen's *square principle* $\square_\kappa$ holds, then any $<\kappa^+$-strategically closed poset has filters meeting κ^+ many dense sets simultaneously.

Ishiu-Yoshinobu improved this result to show that, under the same assumption, any $<\kappa^+$-strategically closed poset is actually κ^+-strategically closed.

Another related property is that of strong distributivity, see [13, Definition 3.1]:

Definition 1.7. Let $\mathbb{P}$ be a forcing notion and κ a cardinal. $\mathbb{P}$ is *strongly $<\kappa$-distributive* if whenever $(D_\alpha)_{\alpha<\kappa}$ is a sequence of open dense subsets of $\mathbb{P}$ and $p \in \mathbb{P}$ there is a descending sequence $(p_\alpha)_{\alpha<\kappa}$ such that $p_0 \leq p$ and $p_\alpha \in D_\alpha$ for any $\alpha < \kappa$.

By [13, Theorem 3.6], this is equivalent to the statement that INC does not have a winning strategy in $G(\mathbb{P}, \kappa)$.

The forcing adding a witness to $\neg F^+(\kappa)$ will always be κ-strategically closed while the forcing adding a witness to $\neg F(\kappa)$ cannot even be *strongly* $<\kappa$-distributive assuming $F(\kappa)$ holds in the ground model (however it will be $<\kappa$-strategically closed and thus $<\kappa$-distributive assuming $F(\delta)$ fails for all $\delta < \kappa$). We will see later in Remark 6.8 that there is a natural separation between $<\kappa$-strategic closure and κ-strategic closure: While the second property suffices for the common task of building sufficiently generic filters, the first property does not.

On the positive side, we have the following ZFC results:

Theorem C. *Let κ be a cardinal.*

(1) *Assume $F(\delta)$ fails for all $\delta < \kappa$.*
   (a) *If κ is weakly compact, $F(\kappa)$ fails.*
   (b) *If $\lambda \geq \kappa$ is such that κ is λ-supercompact, $F(\lambda^+)$ fails.*

(2) *Assume $F^+(\delta)$ fails for all $\delta < \kappa$ and $\lambda \geq \kappa$ is such that κ is λ-supercompact. Then $F^+(\delta)$ fails for all $\delta \leq \lambda$.*

So certain large cardinal properties have a different effect on the patterns of F and F^+ respectively. This is contrasted by the following independence results:

Theorem D. *Assume $F^+(\delta)$ holds for all δ and $\kappa \leq \lambda$ are cardinals such that κ is λ^+-supercompact.*

(1) *There is a forcing extension in which $F(\delta)$ fails for all $\delta \leq \lambda^+$, $F^+(\lambda^{++})$ holds and κ is λ-supercompact.*

(2) *There is a forcing extension in which $F^+(\delta)$ fails for all $\delta \leq \lambda$, $F^+(\lambda^+)$ holds and κ is λ-supercompact.*

1.1. On the consistency of $F^+(\omega_2)$.

In this subsection we give an easier proof of a result due to Shelah that $F^+((D_i)_{i\in\omega_1}, \omega_2)$ is consistent from a Mahlo cardinal, where $(D_i)_{i\in\omega_1}$ is the trivial partition given by $D_0 := \omega_1$. Recall that this principle was fixed as $F^+(\omega_2)$.

There is a very natural forcing which, given some stationary $A \subseteq E^\kappa_\omega$, adds a closed set $c \subseteq A$ with ordertype ω_1:

Definition 1.8. Let κ be a regular cardinal and $A \subseteq E^\kappa_\omega$ stationary. The forcing $\mathbb{P}(A)$ is defined as follows: $p \in \mathbb{P}(A)$ if $p\colon \alpha + 1 \to A$ is a normal function for some $\alpha < \omega_1$. $q \leq p$ in $\mathbb{P}(A)$ if q end-extends p.

In most cases, this poset collapses κ to ω_1 and is not proper (assuming $E^\kappa_\omega \setminus A$ is stationary, as this is destroyed). We do not know if this poset is always semiproper, but it is easy to see that it preserves stationary subsets of ω_1 (we include the original proof from [5] since its ideas will reappear later in the paper):

Lemma 1.9. *Let $\kappa \geq \omega_2$ be regular and $A \subseteq E^\kappa_\omega$ be stationary. Then $\mathbb{P}(A)$ preserves stationary subsets of ω_1.*

Proof. Let $S \subseteq \omega_1$ be stationary and $\dot{C}$ a $\mathbb{P}(A)$-name for a club. Let Θ be large enough and $F\colon [H(\Theta)]^{<\omega} \to [H(\Theta)]^{<\omega_1}$ a function such that any $M \in [H(\Theta)]^{<\omega_1}$

that is closed under F is an elementary submodel of $(H(\Theta), \in)$ containing $\dot{C}$ and $\mathbb{P}(A)$. Let $M \in [H(\Theta)]^{\omega_1}$ be closed under F with $M \cap \omega_2 \in A$. Let $\{\delta_n \mid n \in \omega\} \subseteq M \cap \omega_2$ be unbounded in $\sup(M \cap \omega_2)$. We can find $N \subseteq M$ such that $N \in [H(\Theta)]^{<\omega_1}$ is closed under F, contains $\{\delta_n \mid n \in \omega\}$ as a subset and $N \cap \omega_1 \in S$. Then $\sup(N \cap \omega_2) = \sup(M \cap \omega_2) \in A$, $N \prec (H(\Theta), \in)$ and $\dot{C}, \mathbb{P}(A) \in N$. Let $(D_n)_{n \in \omega}$ enumerate all open dense subsets of $\mathbb{P}(A)$ lying in N. In N, construct a descending sequence $(p_n)_{n \in \omega}$ such that $p_n \in D_n$. It follows that $p := \bigcup_n p_n \cup \{(N \cap \omega_1, \sup(N \cap \omega_2))\} \in \mathbb{P}(A)$ and forces that $\dot{C} \cap (N \cap \omega_1)$ is unbounded in $N \cap \omega_1$, so $N \cap \omega_1 \in S \cap \dot{C}$. □

In [19], chapter XI, Shelah introduces a condition known as the S-condition. This condition holds for $\mathbb{P}(A)$ and is iterable with revised countable support. Moreover, forcing notions satisfying the condition for suitable parameters do not add reals and preserve ω_1. Since it is not the main focus of the paper, we will not state the S-condition here and merely used it as a *black box* to state that a later forcing iteration we construct does not collapse ω_1.

Our aim is to prove the following (see [19, Chapter XI, Theorem 7.1]):

Theorem 1.10. *Suppose κ is a Mahlo cardinal. There is a forcing extension where $\kappa = \omega_2$ and $F^+(\kappa)$ holds.*

As stated before, the principle $F^+((D_i)_{i \in \omega_1}, \omega_2)$ has consistency strength at least a Mahlo cardinal if one D_i is all of ω_1 and consistency strength at least a weakly compact cardinal if at least two D_i are stationary in ω_1. Additionally, if one D_i is club in ω_1 (and thus all other D_i are nonstationary), $F^+((D_i)_{i \in \omega_1}, \omega_2)$ is equivalent to $F^+(\omega_2)$ (see the discussion after Lemma 5.8) and thus, again, has consistency strength at least a Mahlo cardinal.

Shelah proves Theorem 1.10 by using a diamond sequence which guesses potential iterations of forcings of the form $\mathbb{P}(A)$, followed by forcings to destroy those stationary sets which do not have closed subsets of ordertype ω_1. However, this requires building master conditions without much knowledge about the exact iteration and is therefore rather technical, even using the S-condition as a blackbox. We will provide an easier proof by first forcing reflection at a Mahlo cardinal and then constructing the iteration without using a diamond sequence and instead using lottery sums. This idea first appeared in work of Hamkins [9], where he shows that one can make large cardinals indestructible by using iterations of lottery sums of posets. It is unclear whether one can simultaneously force the required combination of $\diamondsuit$ and stationary reflection from merely a Mahlo cardinal (see the open questions in the last section).

The natural large cardinal property implying $\mathrm{SR}(\kappa)$ is κ being *weakly compact*. In fact, in L, the statement $\mathrm{SR}(\kappa)$ is equivalent to κ being weakly compact and the statement that every two stationary subsets of ω_2 reflect simultaneously implies that ω_2 is weakly compact in L (see [18]). However, Harrington and Shelah showed in [10] that $\mathrm{SR}(E^{\omega_2}_{\omega})$ (the maximal amount of reflection possible at ω_2) is consistent from a Mahlo cardinal, by first collapsing it to become ω_2 and then iteratively destroying all non-reflecting stationary subsets of ω_2. As they state, if one does not include the collapse, their proof shows the following:

Theorem 1.11 (Harrington-Shelah)**.** *Suppos κ is Mahlo. There is a forcing extension in which κ remains Mahlo and whenever $S \subseteq \kappa$ is stationary, consisting of singular ordinals, there is $\nu < \kappa$ regular such that $S \cap \nu$ is stationary in ν.*

We refer to the statement that all stationary subsets of $\mathrm{Sing} \cap \kappa$ reflect to a regular cardinal by $\mathrm{Refl}(\kappa)$. The main crux now of the argument is that certain κ-cc. posets preserve stationary reflection in some way, which we prove with an

argument similar to Usuba in [20, Proposition 5.1]. We call a poset $\mathbb{P}$ *nicely κ-cc.* if $\mathbb{P}$ is κ-cc., $\mathbb{P} \in H(\kappa^+)$ and there is a club $C \subseteq [H(\kappa^+)]^{<\kappa}$ such that whenever $M \in C$ and $M \cap \kappa$ is inaccessible, $\mathbb{P} \cap M$ is $M \cap \kappa$-cc.

Lemma 1.12. *Assume κ is Mahlo and* Refl(κ) *holds. Let $\mathbb{R}$ be a poset with the following properties:*

(1) *$\mathbb{R}$ is nicely κ-cc, witnessed by C.*
(2) *$\mathbb{R}$ does not change the cofinality of any inaccessible cardinal to ω.*

Let $\dot{S}$ be an $\mathbb{R}$-name for a stationary subset of E^κ_ω and $A, x \in H(\kappa^+)$. Then there is $M \in C$ with $\dot{S}, x, \mathbb{R} \in M$, $M^{<M\cap\kappa} \subseteq M$ and $r \in \mathbb{R} \cap M$ such that

$$r \Vdash_{\mathbb{R}\cap M} \dot{S} \cap (M \cap \kappa) \text{ is stationary in } (M \cap \kappa)$$

Remark 1.13. We note that the requirements of the lemma imply that $\dot{S} \cap (M \cap \kappa)$ is equivalent to an $\mathbb{R} \cap M$-name: For every $\gamma \in M \cap \kappa$, there is a maximal antichain of conditions deciding $\check{\gamma} \in \dot{S}$ inside M. Since $\mathbb{R} \cap M$ is $M \cap \kappa$-cc, the antichain is a subset of M.

Proof. Assume the parameters are given. Let

$$T := \{\delta \in \kappa \mid \exists r \in \mathbb{R}(r \Vdash \check{\delta} \in \dot{S})\}$$

Then T is stationary in κ and does not contain any inaccessible cardinals. Let $(M_i)_{i<\kappa}$ be an $\in$-increasing and continuous sequence of members of C such that $M_i \cap \kappa \in \kappa$, M_{i+1} is closed under $< M_i \cap \kappa$-sequences and $\dot{S}, x, \mathbb{R} \in M_0$. Then $D := \{i < \kappa \mid M_i \cap \kappa = i\}$ is club in κ. It follows that $T \cap D \cap \nu$ is stationary for some regular cardinal $\nu < \kappa$. In particular, $\nu \in D$ and so $\nu = M_\nu \cap \kappa$. It follows that $M \in C$, $\dot{S}, x, \mathbb{R} \in M$ and $M^{<M\cap\kappa} \subseteq M$.

Assume toward a contradiction that there is no $r \in \mathbb{R} \cap M$ that forces $\dot{S} \cap (M \cap \kappa)$ to be stationary in $M \cap \kappa$, i.e. $1_{\mathbb{R}\cap M}$ forces the negation. Because $\mathbb{R} \cap M$ is $(M \cap \kappa)$-cc, there exists a club $E \subseteq M \cap \kappa$ such that $1_{\mathbb{R}\cap M}$ forces $\check{E} \cap \dot{S}$ to be empty. However, there is an ordinal $\gamma \in T \cap E$, since $T \cap (M \cap \kappa)$ is stationary in $M \cap \kappa$. By elementarity, there is $r \in \mathbb{R} \cap M$ which forces $\check{\gamma} \in \dot{S}$. Therefore, r forces $\check{E} \cap \dot{S}$ to be nonempty, a contradiction. □

We can now prove Theorem 1.10:

Proof. Let κ be a Mahlo cardinal. By Theorem 1.11 we may assume that Refl(κ) holds. Let $\mathbb{P} := (\mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha)_{\alpha<\kappa}$ be an RCS iteration with the following iterands:

(1) Case 1: α is inaccessible and $|\mathbb{P}_\beta| < \alpha$ for every $\beta < \alpha$. It follows that $\mathbb{P}_\alpha$ is α-cc. Let $\{\dot{S}_\alpha \mid \alpha < \delta\}$ (for some δ) enumerate all $\mathbb{P}_\alpha$-names for stationary subsets of E^α_ω. Let $\dot{\mathbb{Q}}_\alpha$ be a $\mathbb{P}_\alpha$-name for the lottery sum of $\{\mathbb{P}(\dot{S}_\alpha) \mid \alpha < \delta\}$, i.e. the disjoint union of all $\mathbb{P}(\dot{S}_\alpha)$ for $\alpha < \delta$.
(2) Case 2: Otherwise. We let $\dot{\mathbb{Q}}_\alpha$ be a $\mathbb{P}_\alpha$-name for $\dot{\mathrm{Coll}}(\aleph_1, 2^{|\mathbb{P}_\alpha|})$.

Using Shelah's book [19, Chapter XI] as a blackbox (modulo the easy statement that lottery sums of forcings with the $\{\aleph_2\}$ have the $\{\aleph_2\}$-condition), we see the following:

Claim. *Whenever $\nu \leq \kappa$ is inaccessible, $\mathbb{P}_\nu$ is ν-cc and preserves $\aleph_1$. Additionally, $\mathbb{P}$ does not add reals.*

In particular, $\mathbb{P}$ forces $\kappa = \omega_2$. We show that $F^+(\omega_2)$ holds after forcing with $\mathbb{P}$. So let $\dot{S}$ be a $\mathbb{P}$-name for a stationary subset of E^κ_ω and $p \in \mathbb{P}$. Working with $\{q \in \mathbb{P} \mid q \leq p\}$ instead if necessary, we may assume that $p = 1_\mathbb{P}$. By Lemma 1.12 there is $M \prec H(\kappa^+)$ containing everything relevant such that $M^{<M\cap\kappa} \subseteq M$ and a condition $r \in \mathbb{P} \cap M$ such that

$$r \Vdash_{\mathbb{R}\cap M} \dot{S} \cap (M \cap \kappa) \text{ is stationary in } M \cap \kappa$$

Let $\nu := M \cap \kappa$. Since $r \in \mathbb{P} \cap M$ and we take direct limits at inaccessible cardinals, $\mathrm{supp}(r) \subseteq M$. Let $\dot{S}$ be a $\mathbb{P}_\nu$-name forced by r to be equal to $\dot{S} \cap (M \cap \kappa)$ and by conditions incompatible with r to be equal to $M \cap \kappa$. In particular, $\dot{S}$ is a $\mathbb{P}_\nu$-name for a stationary subset of ν, $\dot{S} = \dot{S}_\alpha$ for some $\alpha < \delta$. Ergo we can let $r' := r \cup \{(\nu, 1_{\mathbb{P}(\dot{S}_\alpha)})\}$ which is a condition in $\mathbb{P}_{\nu+1}$. This condition clearly forces that, at stage ν, we are forcing with $\mathbb{P}(\dot{S}_\alpha)$. In particular, as a condition in $\mathbb{P}$, it forces that there is a closed copy of ω_1 contained in $\dot{S}_\alpha = \dot{S} \cap (M \cap \kappa)$ and therefore in $\dot{S}$. □

The proof of the consistency of the strongest principle is almost the same. Here we start with a weakly compact cardinal instead for the following reason: For $F^+(\omega_2)$ it was enough that any new stationary subset of ω_2 reflects to an inaccessible cardinal below ω_2. However, for $F^+((D_i)_{i\in\omega_1})$ ($D_i = \{i\}$), we need that, given a sequence $(A_i)_{i\in\omega_1}$ of stationary subsets of ω_2, there is $\nu < \omega_2$ inaccessible at which they all reflect. By work of Magidor (see [18]), this requires the existence of a weakly compact cardinal. The remainder of the argument is entirely the same.

2. Adding Witnesses to the Failure of Friedman Properties

In this section we introduce posets which add witnesses to the failure of various instances of F or F^+ and demonstrate their properties.

Definition 2.1. Let $\kappa \geq \omega_2$ be a regular cardinal.

(1) Let $\lambda \leq \kappa$ be any cardinal. $\mathbb{P}_{F(\lambda)}(\kappa)$ consists of regressive functions $p\colon \delta \to \lambda$ for some ordinal $\delta < \kappa$ such that there is no normal function $g\colon \omega_1 \to \delta$ such that $p \circ g$ is constant.

(2) Let $(D_i)_{i\in\omega_1}$ be a partition of ω_1. $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$ consists of functions $p\colon \delta \times \omega_1 \to 2$ for some ordinal $\delta < \kappa$ such that the following holds:
   (a) For any $\alpha \in \delta$ there is at most one $i \in \omega_1$ such that $p(\alpha, i) = 1$.
   (b) For any $\alpha \in \delta$ and $i \in \omega_1$, if $\mathrm{cf}(\alpha) \neq \omega$, $p(\alpha, i) = 0$.
   (c) There is no normal function $g\colon \omega_1 \to \delta$ such that for all $i \in \omega_1$, $p[g[D_i] \times \{i\}] = \{1\}$.

We order both posets by end-extension.

Remark 2.2. It is easy to see that both posets are countably closed, since a union $\bigcup_{n\in\omega} A_n$ of sets where A_{n+1} end-extends A_n contains a closed copy of ω_1 if and only if one of the A_n contains a closed copy of ω_1.

Remark 2.3. The "disjointness condition" (a) is included in the definition of the poset $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$ for the following reason: Without it, given a $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$-generic filter G and $F := \bigcup G$, the set $A := \{\alpha \in \kappa \mid \forall i \in \omega_1 (F(\alpha, i) = 1)\}$ would be a stationary subset of E^κ_ω not containing any closed copy of ω_1, so actually the weakest form of $F^+((E_i)_{i\in\omega_1}, \kappa)$ would fail in the extension and the poset would be ill-suited to force a distinction between variants of F^+. This is also reflected by our assumptions in Theorem 5.9 where we require our sequence of sets witnessing the negation of F^+ to be pairwise disjoint.

We now show that neither poset collapses cardinals (at least assuming GCH). Since both posets have size $\kappa^{<\kappa} = \kappa$, this will follow from them being $< \kappa$-distributive. For the F^+-poset, we can show that it is κ-strategically closed. For the F-poset, we have to be more careful: If $F(\lambda, \kappa')$ holds for some $\kappa' < \kappa$, then $F(\lambda, \kappa)$ is actually equal to $F(\lambda, \kappa')$ (since it contains no conditions of length $\geq \kappa'$). However, if $F(\lambda, \kappa')$ fails for every $\kappa' < \kappa$, we will be able to show that $F(\lambda, \kappa)$ is $< \kappa$-strategically closed. Assuming $F(\lambda, \kappa)$ holds, this is the best possible.

Lemma 2.4. *Let $\kappa \geq \omega_2$ be a regular cardinal and $(D_i)_{i\in\omega_1}$ any partition of ω_1. $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$ is κ-strategically closed.*

Proof. We construct our strategy in such a way that, at limits of uncountable cofinality, there is a club on which our conditions take value 0 no matter the second argument.

Assume we are given a sequence $(p_\alpha)_{\alpha<\gamma}$ for $\gamma < \kappa$. We let $p'_\gamma := \bigcup_{\alpha<\gamma} p_\alpha$ and p_γ a function with domain $(\mathrm{dom}(p'_\gamma)+1)\times\omega_1$, where $p_\gamma(\mathrm{dom}(p'_\gamma), i) = 0$ for any $i < \omega_1$. We now show that this strategy works. Assume toward a contradiction that g is a normal function from ω_1 to $\mathrm{dom}(p_\gamma)$ such that $p_\gamma[g[D_i]\times\{i\}] = \{1\}$ for any $i \in \omega_1$. It follows that g actually maps to $\mathrm{dom}(p'_\gamma)$. Moreover, $\mathrm{im}(g)$ is unbounded in $\mathrm{dom}(p'_\gamma)$ (therefore it is club), since all the p_α for $\alpha < \gamma$ are conditions. Furthermore, by the construction, the set $\{\mathrm{dom}(p'_\alpha) \mid \alpha < \gamma \text{ is a limit}\}$ is club in $\mathrm{dom}(p'_\gamma)$ as well, so there is some $\beta \in \omega_1$, $\beta \in D_i$ for some $i \in \omega_1$, such that $g(\beta) = \mathrm{dom}(p'_\alpha)$ for some limit $\alpha < \gamma$. However, in that case we have $p_\gamma(\mathrm{dom}(p'_\alpha), i) = p_\alpha(\mathrm{dom}(p'_\alpha), i) = 0$, a contradiction. □

In the case of $F(\lambda,\kappa)$, we cannot expect such a strong result. It is easy to see that the proof of the previous Lemma (or its straighforward modification) would fail for $\mathbb{P}_{F(\lambda)}(\kappa)$ since we also would need to remove the possibility that p_γ is constantly 0 on a closed copy of ω_1. Additionally, F reflects downwards, so assuming $F(\lambda,\kappa)$ holds we cannot force $F(\lambda,\kappa^+)$ to fail without adding a κ-length sequence of ordinals. However, this is the only obstacle:

Lemma 2.5. *Let κ be regular and $\lambda \leq \kappa$ any cardinal. Let $\delta < \kappa$ be regular. The following are equivalent:*

(1) *$\mathbb{P}_{F(\lambda)}(\kappa)$ is $\delta+1$-strategically closed.*
(2) *$\mathbb{P}_{F(\lambda)}(\kappa)$ is $<\delta^+$-distributive.*
(3) *$F(\lambda,\delta)$ fails.*

Remark 2.6. We abuse notation slightly in two ways: We consider the principle $F(\lambda,\delta)$ for $\lambda > \delta$ (in which case it is clearly equivalent to $F(\delta,\delta)$) and also for $\delta = \omega_1$ (in which case it clearly always fails).

Proof. A theorem of Foreman [6] states that a poset $\mathbb{P}$ is $<\delta^+$-distributive if and only if INC does not have a winning strategy in $G(\mathbb{P}, \delta+1)$. This easily shows that (1) implies (2). Additionally, if $F(\lambda,\delta)$ holds, $\mathbb{P}_{F(\lambda)}(\kappa)$ does not contain any conditions of length δ. Ergo INC has an easy winning strategy in $F(\lambda,\delta)$ by simply increasing the domain of the previously played condition by one ordinal. This shows that (2) implies (3).

Now assume $F(\lambda,\delta)$ fails as witnessed by a regressive function $f\colon\delta\to\lambda$. We will construct a winning strategy σ for COM in $G(\mathbb{P},\delta+1)$. Assume $(p_\alpha)_{\alpha<\gamma}$ is a run of the game where $\gamma \leq \delta$ is even. If γ is not a limit, we simply let p_γ be an arbitrary extension of p_β with $\mathrm{dom}(p_\gamma) \geq \gamma$ (by assuming inductively that $\mathrm{dom}(p_{\gamma-2}) \geq \gamma-2$, this is clearly possible without extending the domain by more than 2 elements so we can do this easily without leaving $\mathbb{P}_{F(\lambda)}(\kappa)$). Now assume γ is a limit. Let $p'_\gamma := \bigcup_{\alpha<\gamma} p_\alpha$ and p_γ an extension of p'_γ with domain $\mathrm{dom}(p'_\gamma)+1$ such that $p_\gamma(\mathrm{dom}(p'_\gamma)) = f(\gamma)$ (if $\gamma < \delta$, otherwise arbitrary). We will show that this works as intended. First of all, p_γ is regressive because $\mathrm{dom}(p'_\gamma) \geq \gamma$ and so

$$p_\gamma(\mathrm{dom}(p'_\gamma)) = f(\gamma) < \gamma \leq \mathrm{dom}(p'_\gamma)$$

Secondly, assume $g\colon\omega_1\to\mathrm{dom}(p_\gamma)$ is a normal function such that $p_\gamma\circ g$ is constant. As in the previous proof, $\mathrm{im}(g)$ is club in $\mathrm{dom}(p'_\gamma)$. The set $c := \{\mathrm{dom}(p'_\alpha) \mid \alpha < \gamma \text{ is a limit}\}$ is club in $\mathrm{dom}(p'_\gamma)$ as well and therefore so is $c\cap\mathrm{im}(g) =: d$. It follows that there is a normal and surjective function $h\colon\omega_1\to d$. For any $\alpha<\omega_1$, let $i(\alpha)$

be such that $h(\alpha) = \mathrm{dom}(p'_{i(\alpha)})$. Then clearly i is a normal function from ω_1 into γ. Moreover, for any $\alpha, \alpha' \in \omega_1$, we have

$$f(i(\alpha)) = p_\gamma(\mathrm{dom}(p'_{i(\alpha)})) = p_\gamma(h(\alpha)) = p_\gamma(h(\alpha')) = p_\gamma(\mathrm{dom}(p'_{i(\alpha)})) = f(i(\alpha'))$$

contradicting the fact that f witnesses the failure of $F(\lambda, \delta)$. □

On the other hand, assuming $F(\lambda, \kappa)$ holds, $\mathbb{P}_{F(\lambda)}(\kappa)$ cannot even be strongly $< \kappa$-distributive and thus also not κ-strategically closed:

Lemma 2.7. *Assume that $F(\lambda, \kappa)$ holds and $\mathbb{P}$ is strongly $< \kappa$-distributive. Then $\mathbb{P}$ forces $F(\check{\lambda}, \check{\kappa})$.*

Proof. Assume that some $p \in \mathbb{P}$ forces that $\dot{g}\colon \check{\kappa} \to \check{\lambda}$ is a regressive function. Using the strong distributivity of $\mathbb{P}$, we can find a descending sequence $(p_\alpha)_{\alpha<\kappa}$ of conditions in $\mathbb{P}$ and a function $f\colon \kappa \to \lambda$ in V such that for all $\alpha < \kappa$, $p_\alpha \Vdash \dot{g}(\alpha) = f(\alpha)$ (considering $(D_\alpha)_{\alpha<\kappa}$ where D_α consists of those conditions which decide $\dot{g}(\alpha)$). Clearly, f is regressive, since $\dot{g}$ is forced to be regressive. By $F(\lambda, \kappa)$, there is $h\colon \omega_1 \to \kappa$ normal such that f is constant on $\mathrm{im}(h)$. Let $\gamma := \sup(\mathrm{im}(h))$. Then p_γ forces that $\dot{g}$ is constant on $\mathrm{im}(\check{h})$. Since $\dot{g}$ and p were arbitrary, we are done. □

This highlights a difference between F and SR:

Remark 2.8. In particular, $F(\kappa, \kappa)$ does not imply $\mathrm{SR}(E^\kappa_\omega)$: Assume that $F(\kappa, \kappa)$ holds and force with the standard poset adding a non-reflecting stationary subset of E^κ_ω by initial segments (see e.g. [1, Example 6.5]). This poset forces $\neg\,\mathrm{SR}(E^\kappa_\omega)$ but preserves $F(\kappa, \kappa)$ by the previous lemma.

Now we show that $\mathbb{P}_{F(\lambda)}$ and $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}$ work as intended. For $\mathbb{P}_{F(\lambda)}$ this is easy:

Lemma 2.9. *Let κ be regular and $\lambda \leq \kappa$ any cardinal. Assume $F(\lambda, \delta)$ fails for any regular cardinal $\delta < \kappa$. Then $\mathbb{P}_{F(\lambda)}(\kappa)$ adds a function $f\colon \kappa \to \lambda$ such that there is no normal function $g\colon \omega_1 \to \kappa$ with $f \circ g$ constant.*

Proof. Because $F(\lambda, \delta)$ fails for any regular cardinal $\delta < \kappa$, for any $\alpha < \kappa$, the set of all conditions $p \in \mathbb{P}_{F(\lambda)}(\kappa)$ with $\alpha \in \mathrm{dom}(p)$ is dense in $\mathbb{P}_{F(\lambda)}(\kappa)$. Consequently, if G is $\mathbb{P}_{F(\lambda)}(\kappa)$-generic, $\bigcup G$ is a function from κ into λ and clearly there is no normal function $g\colon \omega_1 \to \kappa$ such that $f \circ g$ is constant because any such g would necessarily be bounded in κ and in the ground model, since $\mathbb{P}_{F(\lambda)}(\kappa)$ is $< \kappa$-distributive. Ergo it would witness that some initial segment of f is not a condition in $\mathbb{P}_{F(\lambda)}(\kappa)$, a clear contradiction. □

For $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}$ we use the well-known idea that "most" generically added sets are stationary:

Lemma 2.10. *Let κ be regular and $(D_i)_{i\in\omega_1}$ a partition of ω_1. $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$ adds a disjoint sequence $(A_i)_{i\in\omega_1}$ of stationary subsets of E^κ_ω such that there is no normal function $g\colon \omega_1 \to \kappa$ with $g[D_i] \subseteq A_i$ for all $i \in \omega_1$.*

Proof. Let G be a $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$-generic filter and let $F := \bigcup G$. For any $i \in \omega_1$, let $A_i := \{\alpha \in \kappa \mid F(\alpha, i) = 1\}$. By the definition, $A_i \subseteq E^\kappa_\omega$. Moreover, there clearly does not exist a function g as stated. The only part left to show is that any A_i is stationary in κ. To this end, let $\dot{C}$ be a $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$-name for a club subset of κ, as forced by some condition $p \in \mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$.

We now construct sequences $(p_n)_{n\in\omega}$ and $(\gamma_n)_{n\in\omega}$ as follows: Let (p_0, γ_0) be any pair such that $p_0 \leq p$ and $p_0 \Vdash \check{\gamma}_0 \in \dot{C}$. Given (p_n, γ_n), let (p_{n+1}, γ_{n+1}) be any pair with $\mathrm{dom}(p_{n+1}) = \delta \times \omega_1$ for some $\delta > \gamma_n$ and $\gamma_{n+1} > \mathrm{dom}(p_n)$ such that

$p_{n+1} \Vdash \check{\gamma}_{n+1} \in \dot{C}$. Lastly, let $p := \bigcup_{n\in\omega} p_n$ and $\gamma := \bigcup_{n\in\omega} \gamma_n$. It follows by closure of $\dot{C}$ that $p \Vdash \check{\gamma} \in \dot{C}$. Moreover, $\mathrm{dom}(p) = \gamma \times \omega_1$. As such, we can let p' be an extension of p such that $\mathrm{dom}(p') = (\gamma+1) \times \omega_1$ and $p'(\gamma, i) = 1$, $p'(\gamma, j) = 0$ for $j < \omega_1, j \neq i$. Because $\mathrm{cf}(\gamma) = \omega$, $p' \in \mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$ and p' clearly forces that $\dot{C} \cap \dot{A}_i$ is nonempty. □

The other difference between $\mathbb{P}_{F(\lambda)}(\kappa)$ (or $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$) and the poset adding a nonreflecting stationary set lies in the following observation:

Observation 2.11. If $G \subseteq \mathbb{P}_{F(\lambda)}(\kappa)$ (or $G \subseteq \mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$) is a generic filter, $\bigcup G$ is a condition in $(\mathbb{P}_{F(\lambda)}(\kappa^+))^{V[G]}$ (or $(\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa^+))^{V[G]}$).

This is what will later allow us to construct master conditions and lift ground-model embeddings.

Our Observation 2.11 is articulated by the following definition which generalizes the usual notion of $<\kappa$-directed closure.

Definition 2.12. Let $\mathbb{P}$ be a forcing order and $\delta < \kappa$ regular cardinals. We say that a subset $D \subseteq \mathbb{P}$ is $<\delta$-directed if whenever $A \subseteq D$ has size $<\delta$, there is $p \in A$ which is a lower bound of A. We say that $D \subseteq \mathbb{P}$ is $<\kappa$-($<\delta$-directed)-closed if whenever $D \subseteq \mathbb{P}$ is $<\delta$-directed and has size $<\kappa$, there is $p \in \mathbb{P}$ which is a lower bound of D.

Clearly, even though $\mathbb{P}_{F(\lambda)}(\kappa)$ (or $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$) is not even $<\omega_2$-closed, we have the following:

Lemma 2.13. *Let $\kappa \geq \omega_2$ be a regular cardinal.*

1. *For any cardinal $\lambda \leq \kappa$, $\mathbb{P}_{F(\lambda)}(\kappa)$ is $<\kappa$-($<\omega_2$)-directed closed.*
2. *For any partition $(D_i)_{i\in\omega_1}$ of ω_1, $\mathbb{P}_{F^+((D_i)_{i\in\omega_1})}(\kappa)$ is $<\kappa$-($<\omega_2$)-directed closed.*

Proof. We just prove (1) for $\lambda = 2$, the rest are the same. So let $D \subseteq \mathbb{P}_{F(2)}(\kappa)$ be $<\omega_2$-directed. In particular, any two conditions in D are compatible and therefore D is linearly ordered by $\leq$. Let $p := \bigcup D$ and assume that there is $g\colon \omega_1 \to \mathrm{dom}(p)$ such that $p \circ g$ is constant. For each $\alpha \in \omega_1$, choose $q_\alpha \in D$ with $g(\alpha) \in \mathrm{dom}(q_\alpha)$. By assumption, there is q which extends every q_α. Clearly, q is an initial segment of p and $\mathrm{dom}(q) \supseteq \mathrm{im}(g)$. Ergo, q is not a condition in $\mathbb{P}_{F(2)}(\kappa)$, a contradiction. □

This is connected to our observation as follows:

Lemma 2.14. *Let δ be a cardinal and $\mathbb{P}$ a poset which is $<\delta$-distributive. Let G be a $\mathbb{P}$-generic filter. Then G is a $<\delta$-directed subset of $\mathbb{P}$ in $V[G]$.*

Proof. Assume $A \subseteq G$ has size $<\delta$. Then in particular, $A \in V$. For any $p \in A$, the set $D_p := \{q \in \mathbb{P} \mid q \perp p \vee q \leq p\}$ is open dense. Ergo, so is the intersection $D := \bigcap_{p\in A} D_p$ which is a member of the ground model. In particular, there is $q \in G \cap D$. Since G is a filter, q cannot be incompatible with any $p \in A \subseteq G$. Therefore, q is a lower bound of A. □

Later on, we will use the following *ad hoc* iteration theorem:

Theorem 2.15. *Let $(\mathbb{P}_i, \dot{\mathbb{Q}}_i)_{i<\gamma}$ be an Easton-support forcing iteration. Let $\delta < \kappa$ be regular cardinals and assume that $\mathbb{P}_i$ is trivial for $i < \kappa$. Then if each $\mathbb{P}_i$ is $<\delta$-distributive and forces that $\dot{\mathbb{Q}}_i$ is $<\check{\kappa}$-($<\check{\delta}$-directed)-closed for every $i < \gamma$, every $\mathbb{P}_i$ is $<\kappa$-($<\delta$-directed)-closed.*

Proof. We prove the statement by induction on γ. Suppose it holds for all $\gamma' < \gamma$ and $(\mathbb{P}_i, \dot{\mathbb{Q}}_i)_{i<\gamma}$ is as in the statement of the theorem. If γ is a limit, the statement automatically holds. Assume $\gamma = \gamma' + 2$. Then we have to show that $\mathbb{P}_{\gamma'+1}$ is $<\kappa$-($<\delta$-directed)-closed. This follows from the following two-step case:

Claim. *Suppose $\mathbb{P}$ is $<\kappa$-($<\delta$-directed)-closed, $<\delta$-distributive and forces that $\dot{\mathbb{Q}}$ is $<\check{\kappa}$-($<\check{\delta}$-directed)-closed. Then $\mathbb{P}*\dot{\mathbb{Q}}$ is $<\kappa$-($<\check{\delta}$-directed)-closed.*

Proof. Suppose that $D \subseteq \mathbb{P}*\dot{\mathbb{Q}}$ is $<\delta$-directed and has size $<\kappa$, i.e. $D = \{(p_\alpha, \dot{q}_\alpha) \mid \alpha < \lambda\}$, $\lambda < \kappa$. In particular, $\{p_\alpha \mid \alpha < \lambda\}$ is $<\delta$-directed, has size $<\kappa$ and thus has a lower bound p'. We claim that p' forces $\{\dot{q}_\alpha \mid \alpha < \lambda\}$ to be $<\delta$-directed. To this end, let $A \subseteq \lambda$ have size $<\delta$. Then $A \in V$ by the distributivity of $\mathbb{P}$. So there is $(p'', \dot{q}) \in D$ which is a lower bound of $\{(p, \dot{q}_\alpha) \mid \alpha \in A\}$. This means that p'' forces that $\dot{q}$ is a lower bound of $\{\dot{q}_\alpha \mid \alpha \in A\}$. Since $p' \leq p''$, p' forces this as well, so we are done. Now simply let $\dot{q}'$ be a lower bound of $\{\dot{q} \mid (p, \dot{q}) \in D\}$. Then $(p', \dot{q}')$ is a lower bound of D and we are finished. □

Now assume $\gamma = \gamma' + 1$, γ' is a limit. We check two cases: First assume that γ' is inaccessible and $|\mathbb{P}_i| < \gamma'$ for every $i < \gamma'$. If $\gamma' \leq \kappa$, we are done since $\mathbb{P}_{\gamma'}$ is trivial. Otherwise, let $D \subseteq \mathbb{P}_{\gamma'}$ be $<\delta$-directed, $|D| < \kappa$. Then $D \subseteq \mathbb{P}_{\gamma''}$ for $\gamma'' < \gamma'$ since $\mathrm{cf}(\gamma') > \kappa$. By the inductive hypothesis we can find a lower bound of D in $\mathbb{P}_{\gamma''}$ which is also a lower bound of D in $\mathbb{P}_{\gamma'}$.

Now assume that γ' is not inaccessible. Therefore, $\mathbb{P}_{\gamma'}$ consists of all functions p on γ' such that $p \restriction \beta \in \mathbb{P}_\beta$ for every $\beta < \gamma'$. Let $D \subseteq \mathbb{P}_{\gamma'}$ be $<\delta$-directed with size $<\kappa$, $D = \{p_\alpha \mid \alpha < \lambda\}$. Write $p_\alpha = (p^\alpha_\beta)_{\beta<\gamma'}$. We construct p such that, for each $\beta < \gamma'$, $p \restriction \beta \in \mathbb{P}_\beta$ and is stronger than all $p_\alpha \restriction \beta$. Assume that $p \restriction \beta$ has been constructed. Then it follows just as before that $p \restriction \beta$ forces that $\{p^\alpha_\beta \mid \alpha < \lambda\}$ is $<\delta$-directed: For $A \subseteq \lambda$, we have $A \in V$ since $\mathbb{P}_\beta$ is $<\delta$-distributive. Therefore there is $\nu \in \lambda$ such that p_ν is a lower bound of $\{p_\alpha \mid \alpha \in A\}$. It follows that $p_\nu \restriction \beta$ forces that p^ν_β is a lower bound of $\{p^\alpha_\beta \mid \alpha \in A\}$. Since $p \restriction \beta \leq p_\nu \restriction \beta$, we are done. Thus, we can find, by the maximum principle, $p(\beta)$ forced to be a lower bound of $\{p^\alpha_\beta \mid \alpha < \lambda\}$. If every p^α_β is trivial, we let $p(\beta)$ be trivial as well.

Now all that is left is to show that $p \restriction \beta \in \mathbb{P}_\beta$ for every β. If β is not inaccessible and $> \kappa$, there is nothing to show, since we are taking the indirect limit. Otherwise, as before, every $p_\alpha \restriction \beta$ is bounded in β and therefore there is $\beta' < \beta$ such that every $p_\alpha \restriction \beta$ is trivial after β'. Ergo, so is $p \restriction \beta$ and $p \restriction \beta \in \mathbb{P}_\beta$. Thus, p is a lower bound of $\{p_\alpha \mid \alpha \in A\}$. □

Propagation of $\neg F(\delta)$.

In this subsection we use the properties of the forcing notion $\mathbb{P}_{F(\lambda)}(\kappa)$ and well-known statements regarding strategic closure in order to show that the existence of witnesses to $\neg F(\lambda, \kappa)$ can be lifted to arbitrary ordinals below κ^+ and to show that the failure of $F(\lambda, \delta)$ is compact at singular cardinals.

The following was first shown by Velleman[21]:

Lemma 2.16. *Assume δ is a singular limit ordinal and $\mathbb{P}$ is a forcing order which is $<\delta$-strategically closed. Then $\mathbb{P}$ is $\delta+1$-strategically closed.*

Proof. By the assumption of the lemma, $\mathbb{P}$ is in particular $\mathrm{cf}(\delta)+1$-strategically closed. Fix a winning strategy σ for COM in $G(\mathbb{P}, \mathrm{cf}(\delta)+1)$ and also a winning strategy σ_α for COM in $G(\mathbb{P}, \alpha+1)$ for any $\alpha < \delta$. For simplicity, we will assume that any such strategy maps $(1_\mathbb{P})_{\beta<\gamma}$ to $1_\mathbb{P}$. We will construct a winning strategy σ_δ for COM in $G(\mathbb{P}, \delta+1)$. To this end, fix a normal function $f\colon \mathrm{cf}(\delta) \to \delta$ that maps successor ordinals to successor ordinals.

Let $(p_\beta)_{\beta<\gamma}$ be given such that $\gamma < \lambda$ is even (so γ may be a limit ordinal).

As a first case, assume $\gamma \notin \mathrm{im}(f)$. Let α be minimal such that $\gamma < f(\alpha+1)$. Then we let

$$\sigma_\delta((p_\beta)_{\beta<\gamma}) := \sigma_{f(\alpha+1)}((1_\mathbb{P})_{\beta\leq f(\alpha)}{}^\frown (p_\beta)_{\beta\in(f(\alpha),\gamma)})$$

This is possible because, by induction, $(1_{\mathbb{P}})_{\beta<f(\alpha)}{}^\frown(p_\beta)_{\beta\in(f(\alpha),\gamma)}$ has been played according to $\sigma_{f(\alpha+1)}$, since $\sigma_{f(\alpha+1)}$ maps the constant $1_{\mathbb{P}}$-sequence to $1_{\mathbb{P}}$ by assumption and the first nonzero condition is $p_{f(\alpha)+1}$ after which we used $\sigma_{f(\alpha+1)}$ (noting that INC moved at $f(\alpha)+1$). Additionally, $\sigma_\delta((p_\beta)_{\beta<\gamma})$ is a lower bound of $(p_\beta)_{\beta\in(f(\alpha),\gamma)}$ and thus of $(p_\beta)_{\beta<\gamma}$ as required.

Now assume $\gamma = f(\alpha)$ for some α. If $\alpha = \xi+1$, we can find μ such that $\gamma = \mu+1$ and we let

$$\sigma_\delta((p_\beta)_{\beta<\gamma}) := \sigma((p_{f(\beta)})_{\beta<\xi}{}^\frown p_\mu)$$

this is a lower bound of $(p_\beta)_{\beta<\gamma}$ because $\gamma = \mu+1$.

If α is a limit ordinal, we let

$$\sigma_\delta((p_\beta)_{\beta<f(\alpha)}) := \sigma((p_{f(\beta)})_{\beta<\alpha})$$

this works because $(p_{f(\beta)})_{\beta<\alpha}$ has been played according to σ.

Lastly, if $\gamma = \delta$, then $\sigma_\delta((p_\beta)_{\beta<\gamma}) = \sigma((p_{f(\beta)})_{\beta<\mathrm{cf}(\delta)})$. Ergo σ_δ is as required. □

We obtain an easy corollary which is a well-known folklore result.

Corollary 2.17. *Let $\mathbb{P}$ be a forcing order and μ a cardinal. If $\mathbb{P}$ is $\mu+1$-strategically closed, $\mathbb{P}$ is $<\mu^+$-strategically closed.*

Proof. One shows easily by induction that for any δ, $\mathbb{P}$ is $\delta+1$-strategically closed. The successor step is clear, in limits we use that any $\delta \in (\mu, \mu^+)$ is singular. □

By relating the strategic closure of $\mathbb{P}_{F(\lambda)}(\kappa)$ to the failure of $F(\lambda,\kappa)$, we see the following:

Lemma 2.18. *Let $\kappa \geq \omega_2$ be regular and $\lambda \leq \kappa$ any cardinal. Assume $F(\lambda,\kappa)$ fails and $\delta \in [\kappa, \kappa^+)$.*

1. *If $\lambda < \kappa$ there is a regressive function $g\colon \delta \to \lambda$ that is not constant on any closed copy of ω_1.*
2. *If $\lambda = \kappa$ there is a regressive function $g\colon \delta \to \delta$ that is not constant on any closed copy of ω_1.*

Proof. Assume $F(\lambda,\kappa)$ fails. Then $\mathbb{P}_{F(\lambda)}(\kappa^+)$ is $\kappa+1$-strategically closed by Lemma 2.5. By Corollary 2.17, $\mathbb{P}_{F(\lambda)}(\kappa^+)$ is $\delta+1$-strategically closed. Ergo $\mathbb{P}_{F(\lambda)}(\kappa^+)$ contains a condition with domain δ which is as required. □

A similar proof shows that the negation of F is compact at singular cardinals (recall Remark 2.6):

Lemma 2.19. *Let κ be singular and $\lambda \leq \kappa$ any cardinal. Assume $F(\lambda,\delta)$ fails for all $\delta < \kappa$.*

1. *If $\lambda < \kappa$, there is a regressive function $g\colon \kappa \to \lambda$ that is not constant on any closed copy of ω_1.*
2. *If $\lambda = \kappa$, there is a regressive function $g\colon \kappa \to \kappa$ that is not constant on any closed copy of ω_1.*

Proof. Assuming $F(\lambda,\delta)$ fails for all $\delta < \kappa$, the poset $\mathbb{P}_{F(\lambda)}(\kappa^+)$ is $<\kappa$-strategically closed. Ergo it is actually $\kappa+1$-strategically closed. It follows that functions as required exist. □

3. Maximal Versions of MM with Failure of Friedman Properties

The forcing axiom Martin's Maximum (MM) was introduced by Foreman, Magidor and Shelah in [5]. It is shown in the same paper that MM implies $F^+((D_i)_{i\in\omega_1},\kappa)$ for any regular $\kappa \geq \omega_2$, where $(D_i)_{i\in\omega_1}$ is the partition given by $D_0 = \omega_1$, $D_i = \emptyset$ for $i \neq 0$ (we will later show that SRP, a consequence of MM, implies $F^+((D_i)_{i\in\omega_1},\kappa)$

for any partition $(D_i)_{i\in\omega_1}$ and any regular $\kappa \geq \omega_2$). In this section we will introduce variants of MM which are compatible with the failure of various instances of F and F^+ and provably maximal in this respect. The principal applications of these axioms will come in the next two sections where we use them to separate all instances of the defined principles. We will focus on the failure of the principles at ω_2 to avoid making the arguments and notation too cumbersome.

Definition 3.1. Let $\kappa \geq \omega_2$ be regular.

(1) Let $\lambda \leq \kappa$ be any cardinal and $f\colon \kappa \to \lambda$ regressive. $F_\neg(f, \lambda, \kappa)$ states that f is not constant on any closed copy of ω_1.

(2) Let $(D_i)_{i\in\omega_1}$ be a partition of ω_1 and $(A_i)_{i\in\omega_1}$ a sequence of stationary subsets of E^κ_ω. $F^+_\neg((A_i)_{i\in\omega_1}, (D_i)_{i\in\omega_1}, \kappa)$ states that there is no normal function $g\colon \omega_1 \to \kappa$ such that $g[D_i] \subseteq A_i$.

So, roughly speaking, the previous definition states that f (or $(A_i)_{i\in\omega_1}$) witnesses that the corresponding principle fails.

We can now define our versions of MM:

Definition 3.2. (1) Let $\lambda \leq \omega_2$ be a cardinal and $f\colon \omega_2 \to \lambda$ regressive. $\mathrm{MM}_{F(\lambda)}(f)$ states the following:

- $F_\neg(f, \lambda, \omega_2)$ holds.
- Whenever $\mathbb{P}$ is a poset such that $\mathbb{P}$ preserves stationary subsets of ω_1 as well as $F_\neg(f, \lambda, \omega_2)$ and $\mathcal{D}$ is an ω_1-sized collection of dense subsets of $\mathbb{P}$ there is a filter $G \subseteq \mathbb{P}$ which has nonempty intersection with any $D \in \mathcal{D}$.

(2) Let $(D_i)_{i\in\omega_1}$ be a partition of ω_1 and $(A_i)_{i\in\omega_1}$ a sequence of stationary subsets of $E^{\omega_2}_\omega$. $\mathrm{MM}_{F^+((D_i)_{i\in\omega_1})}((A_i)_{i\in\omega_1})$ states the following:

- $F^+_\neg((A_i)_{i\in\omega_1}, (D_i)_{i\in\omega_1}, \kappa)$ holds.
- Whenever $\mathbb{P}$ is a poset such that $\mathbb{P}$ preserves stationary subsets of ω_1 as well as $F^+_\neg((A_i)_{i\in\omega_1}, (D_i)_{i\in\omega_1}, \omega_2)$ and $\mathcal{D}$ is an ω_1-sized collection of dense subsets of $\mathbb{P}$ there is a filter $G \subseteq \mathbb{P}$ which has nonempty intersection with any $D \in \mathcal{D}$.

The main result of this section is that any such principle is consistent. Thanks to Observation 2.11 showing this is surprisingly straightfoward: We obtain a model of any such principle by forcing over the original model of Martin's Maximum from [5] with the poset adding a witness to the failure of the corresponding property.

We first give some background on Martin's Maximum. The notions of *proper* and *semiproper forcing* were introduced by Shelah (see e.g. his famous book [19]). Recall that for any poset $\mathbb{P}$, any set M and any $\mathbb{P}$-generic filter G, we let $M[G] := \{\tau^G \mid \tau \in M \text{ is a } \mathbb{P}\text{-name}\}$.

Definition 3.3. Let $\mathbb{P}$ be a poset, $M \prec H(\Theta)$ with $\mathbb{P} \in M$ and $p \in \mathbb{P}$.

(1) p is $(M, \mathbb{P})$-*generic* if $p \Vdash M[\Gamma] \cap V = M$.

(2) p is $(M, \mathbb{P})$-*semigeneric* if $p \Vdash M[\Gamma] \cap \omega_1 = M \cap \omega_1$.

$\mathbb{P}$ is *proper* (*semiproper*) if for any sufficiently large cardinal Θ there is a club $C \subseteq [H(\Theta)]^{<\omega_1}$ such that whenever $M \in C$ and $p \in M \cap \mathbb{P}$ there is an $(M, \mathbb{P})$-generic ($(M, \mathbb{P})$-semigeneric) condition $q \leq p$.

It can be shown that any proper notion of forcing preserves stationary subsets of $[X]^{<\omega_1}$ for all sets X and any semiproper notion of forcing preserves stationary subsets of ω_1. Moreover, properness and semiproperness is preserved by iterations with suitable support: For proper forcing, countable support suffices while semiproperness is preserved by iterations with *revised countable support* (as stated in Section 1, we will not go into that definition here).

The other ingredient we need is the concept of a *Laver function* from [17]:

Definition 3.4. Let κ be a supercompact cardinal. $f\colon \kappa \to V_\kappa$ is a *Laver function* if for any set x and any $\lambda \geq |\mathrm{tcl}(x)|$ there is an elementary embedding $j\colon V \to M$, where M is transitive, such that $j(\kappa) > \lambda$, ${}^{\lambda}M \subseteq M$ and $j(f)(\kappa) = x$.

Lastly, we need that certain notions of forcing preserve witnesses to the failure of F (or F^+).

We start with the following well-known result:

Lemma 3.5. *Let γ be an ordinal with $\mathrm{cf}(\gamma) > \omega$ and $A \subseteq E^\gamma_\omega$ stationary. Let $\mathbb{P}$ be a poset.*

1. *If $\mathbb{P}$ is proper, A is stationary in $V[\mathbb{P}]$.*
2. *If $\mathbb{P}$ is semiproper and $|\gamma| = \omega_1$ in V, A is stationary in $V[\mathbb{P}]$.*

Proof. Assume toward a contradiction that $\dot{C}$ is a $\mathbb{P}$-name for a club such that some $p \in \mathbb{P}$ forces $\dot{C} \cap \check{A} = \emptyset$. In V, let $D \subseteq [H(\Theta)]^{<\omega_1}$ consist of those $M \prec (H(\Theta), \in, \dot{C})$ such that $\gamma, \mathbb{P}, p \in M$ and M witnesses the (semi-) properness of $\mathbb{P}$. D is club. Ergo there is $M \in D$ such that $\sup(M \cap \gamma) \in A$. Let $q \leq p$ be $(M, \mathbb{P})$-(semi-)generic. Let G be $\mathbb{P}$-generic containing q. In $V[G]$, $M[G]$ is a limit point of $\dot{C}^G$ and thus $\sup(M[G] \cap \gamma) \in \dot{C}^G$. If $\mathbb{P}$ is proper we have $M[G] \cap \gamma = M \cap \gamma$ and obtain a contradiction.

If $\mathbb{P}$ is semiproper and $|\gamma| = \omega_1$, M contains a bijection f between ω_1 and γ (and $M[G]$ thus does as well). By elementarity, $f \restriction (M \cap \omega_1)$ is a bijection between $M \cap \omega_1$ and $M \cap \gamma$ and $f \restriction (M[G] \cap \omega_1)$ is a bijection between $M[G] \cap \omega_1$ and $M[G] \cap \gamma$. Because $M[G] \cap \omega_1 = M \cap \omega_1$, $M[G] \cap \gamma = M \cap \gamma$ and we proceed as above. □

Lemma 3.6. *Let γ be an ordinal with $\mathrm{cf}(\gamma) > \omega$ and $A \subseteq \gamma$ such that A does not contain a closed copy of ω_1.*

1. *If $\mathbb{P}$ is proper, A does not contain a closed copy of ω_1 in $V[\mathbb{P}]$.*
2. *If $\mathbb{P}$ is semiproper and $|\gamma| = \omega_1$ in V, A does not contain a closed copy of ω_1 in $V[\mathbb{P}]$.*

Proof. We prove both parts almost simultaneously.

Assume toward a contradiction that there is a $\mathbb{P}$-name $\dot{f}$ such that some $p \in \mathbb{P}$ forces $\dot{f}$ to be a normal function from ω_1 into $\check{A}$. Assume p decides $\sup(\mathrm{im}(\dot{f})) = \check{\beta}$ for some β (we note that $\beta = \gamma$ is possible). We have $\mathrm{cf}(\beta) = \omega_1$ in $V[\mathbb{P}]$ and thus $\mathrm{cf}(\beta) > \omega$ in V. However, since A does not contain a closed copy of ω_1 in V, $E^\beta_\omega \setminus A$ is stationary in β in V, but this stationarity is destroyed in $V[\mathbb{P}]$. In any case, we obtain a contradiction using Lemma 3.5. □

Interestingly, we need that γ has size ω_1 in V and not in $V[G]$: For any stationary $A \subseteq E^{\omega_2}_\omega$, there is a (at least consistently) semiproper poset that adds a closed copy of ω_1 contained in A while collapsing ω_2 to ω_1.

We can now show the consistency of our versions of MM. By the *standard iteration to force Martin's Maximum* $\mathbb{P}_{MM}$ we mean the following: Let κ be supercompact and $f\colon \kappa \to V_\kappa$ a Laver function. The existence of this function follows from ZFC by Laver's classical paper [17]. Let $(\mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha)_{\alpha < \kappa}$ be a revised countable support iteration such that $\dot{\mathbb{Q}}_\alpha = f(\alpha)$ whenever α is inaccessible and $f(\alpha)$ is a $\mathbb{P}_\alpha$-name for a semiproper poset and $\dot{\mathbb{Q}}_\alpha$ is the Levy collapse of $2^{|\mathbb{P}_\alpha|}$ to ω_1 otherwise.

We first do the proof for any instance of $\mathrm{MM}_{F(\lambda)}(f)$.

Theorem 3.7. *Let κ be a supercompact cardinal and $\lambda \in \omega_1 + 1 \cup \{\kappa\}$. There is a forcing extension where $\kappa = \omega_2$ and there is a function $f\colon \kappa \to \lambda$ such that $\mathrm{MM}_{F(\lambda)}(f)$ holds.*

Proof. Let $\mathbb{P} := \mathbb{P}_{MM} * \dot{\mathbb{P}}_{F(\lambda)}(\dot{\omega}_2)$, where $\mathbb{P}_{MM}$ is the standard iteration to force Martin's Maximum (as above). We will verify that $\mathbb{P}$ gives us the required model.

The first step is showing that after forcing with $\mathbb{P}$, any stationary set preserving forcing is semiproper. This is done using a similar proof as for Lemma 3 in [5]. The idea is that in $V[\mathbb{P}]$ (just like in $V[\mathbb{P}_{MM}]$) any ground-model supercompactness embedding $j: V \to M$ can be extended to $V[\mathbb{P}] \to M[j(\mathbb{P})]$ in an extension of $V[\mathbb{P}]$ using a semiproper poset.

So let $\dot{\mathbb{Q}}$ be a $\mathbb{P}$-name for a forcing notion which preserves stationary subsets of ω_1 but is not semiproper. Let G be any $\mathbb{P}$-generic filter. By the assumption there is a stationary set $A \subseteq [H^{V[G]}(\Theta)]^{<\omega_1}$ (for Θ sufficiently large) such that for any $N \in A$ there is some $p \in N \cap \mathbb{P}$ such that there is no $(N, \mathbb{P})$-semigeneric condition below p. By the normality of the club filter we can assume that there is one such p for all $N \in A$. Assume for simplicity that $p = 1_{\mathbb{Q}}$ and let $\gamma := |H^{V[G]}(\Theta)|$.

In V, let $j: V \to M$ be any γ^+-supercompact embedding (i.e. $j(\kappa) > \gamma^+$ and ${}^{\gamma^+}M \subseteq M$) such that $j(l)(\kappa)$ is a $\mathbb{P}_\kappa$-name for $\mathbb{P}_{F(\lambda)}(\dot{\omega}_2) * \dot{\mathrm{Coll}}(\omega_1, \gamma)$. Because $\mathbb{P}_{F(\lambda)}(\dot{\omega}_2) * \dot{\mathrm{Coll}}(\omega_1, \gamma)$ is countably closed, it is even proper and therefore $j(\mathbb{P}_{MM})$ is isomorphic to $\mathbb{P} * \mathbb{P}_{F(\lambda)}(\dot{\omega}_2) * \dot{\mathrm{Coll}}(\omega_1, \gamma) * \dot{\mathbb{R}}$, where $\dot{\mathbb{R}}$ is forced to be a revised countable support iteration of semiproper forcings (and thus semiproper itself).

Now we show that j lifts to an embedding of $V[G]$. First of all, let G_{MM} be the $\mathbb{P}_{MM}$-generic filter induced by G. Let H_{MM} be any $j(\mathbb{P}_{MM})$-generic filter containing G. Then j lifts to $j^+: V[G_{MM}] \to M[H_{MM}]$. In $M[H_{MM}]$, let q be the union of $G_{F(\lambda)}$, where $G_{F(\lambda)}$ is the $\mathbb{P}_{F(\lambda)}(\dot{\omega}_2)^{G_{MM}}$-generic filter induced by H_{MM}. So $G = G_{MM} * G_{F(\lambda)}$. In $V[G_{MM} * G_{F(\lambda)}]$ (and in $M[G_{MM} * G_{F(\lambda)}]$), q is not constant on any closed copy of ω_1.

We will now show that this is still the case in $M[H_{MM}]$. First of all, let H_{Coll} be the $\dot{\mathrm{Coll}}(\omega_1, \gamma)^{G_{MM} * G_{F(\lambda)}}$-generic filter induced by H_{MM}. Because $\dot{\mathrm{Coll}}(\omega_1, \gamma)^{G_{MM} * G_{F(\lambda)}}$ is countably closed and hence proper, q is still not constant on any closed copy of ω_1 in $M[G_{MM} * G_{F(\lambda)} * H_{\mathrm{Coll}}]$ (by Lemma 3.6 (1)). Moreover, after forcing with $\mathrm{Coll}(\omega_1, \gamma)^{G_{MM} * G_{F(\lambda)}}$, q is a function on an ordinal in ω_2 with cofinality ω_1. Thus, letting $H_{\mathbb{R}}$ be the $\dot{\mathbb{R}}^{G_{MM} * G_{F(\lambda)} * H_{\mathrm{Coll}}}$-generic filter induced by H, q is still not constant on a closed copy of ω_1 in $M[G_{MM} * G_{F(\lambda)} * H_{\mathrm{Coll}} * H_{\mathbb{R}}] = M[H_{MM}]$ because $\dot{\mathbb{R}}^{G_{MM} * G_{F(\lambda)} * H_{\mathrm{Coll}}}$ is semiproper.

In summary, q is a condition in $\mathbb{P}_{F(\lambda)}(\dot{\omega}_2)^{H_{MM}}$. Ergo, letting $H_{F(\lambda)}$ be any $\mathbb{P}_{F(\lambda)}(\dot{\omega}_2)^{H_{MM}}$-generic filter containing q, $j[G_{MM} * G_{F(\lambda)}] \subseteq H_{MM} * H_{F(\lambda)}$ and so j lifts to $j^{++}: V[G_{MM} * G_{F(\lambda)}] \to M[H_{MM} * H_{F(\lambda)}]$. Let $H := H_{MM} * H_{F(\lambda)}$ (recall $G = G_{MM} * G_{F(\lambda)}$), so $j^{++}: V[G] \to M[H]$.

In $M[H]$, A is still stationary in $[H^{V[G]}(\Theta)]^{<\omega_1}$: A is stationary in $M[G]$. $M[H_{MM}]$ has been obtained by forcing with $(\dot{\mathrm{Coll}}(\omega_1, \gamma) * \dot{\mathbb{R}})^G$ over $M[G]$. $\dot{\mathrm{Coll}}(\omega_1, \gamma)^G$ is proper, so it preserves the stationarity of A and makes A (morally) a stationary subset of ω_1, so its stationarity is preserved by the semiproper forcing $\dot{\mathbb{R}}^{G * H_{\mathrm{Coll}}}$. Lastly, A is stationary in $M[H]$ by the countable closure of $\mathbb{P}_{F(\lambda)}(\dot{\omega}_2)^{H_{MM}}$.

By elementarity, $j(\dot{\mathbb{Q}})^H$ preserves stationary subsets of ω_1 in $M[H]$. Thus, if I is $j(\dot{\mathbb{Q}})^H$-generic, A is stationary in $[H(\Theta)^{V[G]}]^{<\omega_1}$ in $M[H * I]$.

In $M[H * I]$, there is a club set $C \subseteq [H(j(\Theta))^{M[H*I]}]^{<\omega_1}$ such that whenever $N \in C$ and $\tau \in N \cap H(\Theta)^{V[G]}$ is a $\dot{\mathbb{Q}}^G$-name for an element of ω_1, then $j^{++}(\tau)^I \in N$. This club set is unrelated to the semi-properness and is simply the club set of all structures closed under the function mapping τ to $j^{++}(\tau)^I$. Ergo there is $N \in C$ such that $N \cap H(\Theta)^{V[G]} \in A$. In particular, since $A \in V[G]$, $N \cap H(\Theta)^{V[G]} \in V[G]$. In $M[H]$, let $q \in j(\mathbb{Q})^H$ force that such an N with $N' := N \cap H(\Theta)^{V[G]} \in A$ exists (for some fixed N'). Then q forces that whenever $\tau \in N'$ is a $\dot{\mathbb{Q}}^G$-name for an element of ω_1, then $j^{++}(\tau)^I \in N'$. Since $N' \in V[G]$, we can view $j^{++}(N')$

and, since N' is countable, $j^{++}(N') = j^{++}[N']$. Ergo q forces that whenever $\tau \in j^{++}(N')$ is a $j(\dot{\mathbb{Q}})^H$-name for an element of ω_1, then $\tau^I \in j(N')$. So in $M[H]$ there exists a semigeneric condition for $j^{++}(N')$ and thus in $V[G]$ there exists a semigeneric condition for N'. This contradicts our assumption.

A very similar argument shows that $\mathrm{MM}_{F(\lambda)}(f)$ holds in $V[G]$: Let $\dot{\mathbb{Q}}$ be a $\mathbb{P}$-name for a forcing that preserves stationary subsets of ω_1 and forces $F(f, \lambda, \omega_2)$ and $\dot{\mathcal{D}}$ a $\mathbb{P}$-name for an ω_1-sized collection of dense subsets of $\dot{\mathbb{Q}}$. Let $j\colon V \to M$ be a γ-supercompact embedding (where $\gamma \geq |\dot{\mathbb{Q}}|$) such that $j(l)(\kappa)$ is equal to $\mathbb{P}_{F(\lambda)}(\omega_2) * \dot{\mathbb{Q}} * \dot{\mathrm{Coll}}(\omega_1, \dot{\omega}_2)$. Just as in the last proof, there is a $j(\mathbb{P})$-generic filter H such that j lifts to $j^{++}\colon V[G] \to M[H]$: $\dot{\mathbb{Q}}$ preserves that the generic function added by $\mathbb{P}_{F(\lambda)}(\omega_2)$ is not constant on a closed copy of ω_1 by assumption, this is further preserved by $\mathrm{Coll}(\omega_1, \omega_2)$ by the properness and lastly by the tail of the iteration as before. In $M[H]$ there is a filter L which is generic for $\dot{\mathbb{Q}}^G$ over $M[G]$. Clearly this filter intersects every element of $\dot{\mathcal{D}}^G$. Moreover, $j^{++}(\dot{\mathcal{D}}^G) = j^{++}[\dot{\mathcal{D}}^G]$ because it has size ω_1 in $V[G]$. Ergo, $j^{++}[L]$ (which is in $M[H]$ by the closure) intersects every element of $j^{++}(\dot{\mathcal{D}}^G)$. By elementarity there is a filter in $V[G]$ for the poset $\dot{\mathbb{Q}}^G$ which intersects every element of $\dot{\mathcal{D}}^G$. Thus $\mathrm{MM}_{F(\lambda)}(f)$ holds. □

We now turn to F^+. We first have the following analogue of Lemma 3.6:

Lemma 3.8. *Let γ be an ordinal with $\mathrm{cf}(\gamma) > \omega$. Let $(D_i)_{i\in\omega_1}$ be a partition of ω_1, $(A_i)_{i\in\omega_1}$ a sequence of stationary subsets of E^γ_ω and $\mathbb{P}$ a poset. Assume there is no normal function $g\colon \omega_1 \to \gamma$ such that $g[D_i] \subseteq A_i$ for all $i \in \omega_1$.*

(1) *If $\mathbb{P}$ is proper, this is preserved by $\mathbb{P}$.*
(2) *If $\mathbb{P}$ is semiproper and $|\gamma| = \omega_1$, this is preserved by $\mathbb{P}$.*

Proof. Assume toward a contradiction that there is $p \in \mathbb{P}$ which forces some $\mathbb{P}$-name $\dot{g}$ to be a normal function from ω_1 to γ such that $\dot{g}[D_i] \subseteq A_i$ for all $i \in \omega_1$. Assume also that p forces $\sup(\mathrm{im}(\dot{g})) = \check{\delta}$ for some $\delta \leq \gamma$ (as before, $\delta = \gamma$ is possible). Denote $\overline{A} := (A_i)_{i\in\omega_1}$ and $\overline{D} := (D_i)_{i\in\omega_1}$.

Let $(M_i)_{i\in\omega_1}$ be an $\in$-increasing and continuous sequence of elements of $[H(\Theta)]^{<\omega_1}$ such that $M_i \prec (H(\Theta), \in)$, $\dot{g}, p, \delta, \gamma, \overline{A}, \overline{D} \in M_i$ and M_i witnesses (semi-)properness for any $i \in \omega_1$.

Claim. *There is $i, j \in \omega_1$ such that $M_i \cap \omega_1 \in D_j$ and $\sup(M_i \cap \delta) \notin A_j$.*

Proof. Otherwise, we could define a normal function $f\colon \omega_1 \to \gamma$ by letting $f(M_i \cap \omega_1) := \sup(M_i \cap \delta)$ for $i \in \omega_1$ and "filling in the gaps" as in the proof of Lemma 5.2. □

So let $i \in \omega_1$ be such that $M_i \cap \omega_1 \in D_j$ and $\sup(M_i \cap \delta) \notin A_j$. Let $q \leq p$ be $(M, \mathbb{P})$-(semi-)generic and let G be a $\mathbb{P}$-generic filter containing q. We now work in $V[G]$. Since $\dot{g}^G \in M_i[G]$ and $M_i[G] \prec (H^{V[G]}(\Theta), \in)$, the image of $M_i[G] \cap \omega_1$ under $\dot{g}^G$ is unbounded in $M_i[G] \cap \delta$, so by normality $\dot{g}^G(M_i[G] \cap \omega_1) = \sup(M_i[G] \cap \delta)$. In any case, we obtain that $M_i[G] \cap \omega_1 = M_i \cap \omega_1$ and $M_i[G] \cap \delta = M_i \cap \delta$ and thus a contradiction. □

Using Lemma 3.8 in place of Lemma 3.6 in the proof of Theorem 3.7, one can see that the following holds:

Theorem 3.9. *Let κ be a supercompact cardinal and $\mathbb{P}_{MM}$ the standard iteration to forced Martin's Maximum defined from κ. In $V[\mathbb{P}_{MM}]$, let $(D_i)_{i\in\omega_1}$ be any partition of ω_1. Then there is a forcing extension of $V[\mathbb{P}_{MM}]$ where there is a sequence $(A_i)_{i\in\omega_1}$ of stationary subsets of $E^{\omega_2}_\omega$ such that $\mathrm{MM}^{F^+((D_i)_{i\in\omega_1})}((A_i)_{i\in\omega_1})$ holds.*

Proof. Simply force with $\mathbb{P}_{MM} * \mathbb{P}_{F^+((\dot{D}_i)_{i\in\omega_1})}(\dot{\omega}_2)$. The main obstacle is showing that a ground-model supercompact embedding $j: V \to M$ lifts to the extension after $\mathbb{P}_{MM} * \mathbb{P}_{F^+((\dot{D}_i)_{i\in\omega_1})}(\dot{\omega}_2)$. So let $j: V \to M$ be sufficiently supercompact such that $j(l)(\kappa)$ is a $\mathbb{P}_{MM}$-name for $\mathbb{P}_{F^+((\dot{D}_i)_{i\in\omega_1})}(\dot{\omega}_2) * \dot{\mathrm{Coll}}(\omega_1, \gamma)$ for γ sufficiently large and let $G = G_{MM} * G_{F((D_i)_{i\in\omega_1})}$ be generic.

It follows as before that $j: V \to M$ lifts to $j^+: V[G_{MM}] \to M[H_{MM}]$ for some H_{MM}-generic filter containing $G_{MM} * G_{F((D_i)_{i\in\omega_1})}$. In $M[H_{MM}]$, let $q := \bigcup G_{F((D_i)_{i\in\omega_1})}$. It then follows as in the proof of Theorem 3.7, replacing Lemma 3.6 by Lemma 3.8 that in $M[H_{MM}]$ there is no normal function g from ω_1 into ω_2 such that $q[g[D_i] \times \{i\}] = \{1\}$, since it is true in $M[G_{MM} * G_{F((D_i)_{i\in\omega_1})}]$ and then preserved by the proper Levy-collapse and afterwards by the semiproper tail. Ergo q is a condition in $\mathbb{P}_{F((D_i)_{i\in\omega_1})}(\omega_2)^{M[H_{MM}]}$ and, taking any generic filter $H_{F^+((D_i)_{i\in\omega_1})}$ containing q, we see that $j^+: V[G_{MM}] \to M[H_{MM}]$ once again lifts to $j^{++}: V[G_{MM} * G_{F^+((D_i)_{i\in\omega_1})}] \to M[H_{MM} * H_{F^+((D_i)_{i\in\omega_1})}]$. The rest of the argument is the same. □

4. Separating Instances of F

We now use $\mathrm{MM}_{F(\lambda)}(f)$ in order to separate all possible instances of $F(\cdot, \omega_2)$.

Theorem 4.1. *Assume $\lambda \leq \omega_2$ is a cardinal and $f: \omega_2 \to \lambda$ is a regressive function such that $\mathrm{MM}_{F(\lambda)}(f)$ holds and $f^{-1}[\{i\}] \cap E^{\omega_2}_\omega$ is stationary in ω_2 for any $i \in \lambda$. Then $F(\lambda', \omega_2)$ holds for all cardinals $\lambda' < \lambda$.*

Proof. Let $\lambda' < \lambda$ be a cardinal and $g: \omega_2 \to \lambda'$ a regressive function.

Claim. *There are $i < \lambda'$ and $j, j' < \lambda$ such that both $g^{-1}[\{i\}] \cap f^{-1}[\{j\}] \cap E^{\omega_2}_\omega$ and $g^{-1}[\{i\}] \cap f^{-1}[\{j'\}] \cap E^{\omega_2}_\omega$ are stationary in ω_2.*

Proof. Let $j < \lambda$. Because $\omega_2 = \bigcup_{i<\lambda'} g^{-1}[\{i\}]$ and $\lambda' < \lambda \leq \omega_2$, there is $h(j) < \lambda'$ such that $g^{-1}[\{h(j)\}] \cap f^{-1}[\{j\}] \cap E^{\omega_2}_\omega$ is stationary. By the pigeonhole principle there are $j, j' < \lambda$ with $h(j) = h(j')$. We can take $i := h(j) = h(j')$. □

Let $\mathbb{P} := \mathbb{P}(g^{-1}[\{i\}] \cap E^{\omega_2}_\omega)$, i.e. the forcing shooting a closed copy of ω_1 into $g^{-1}[\{i\}] \cap E^{\omega_2}_\omega$.

Claim. *$\mathbb{P}$ forces $F(f, \lambda, \omega_2)$.*

Proof. Let $\dot{h}$ be a $\mathbb{P}$-name for a normal function from ω_1 into ω_2 and $p \in \mathbb{P}$. We will show that there is a condition $q \leq p$ that forces $\check{f}(\dot{h}(\check{\alpha})) = \check{j}$ for some $\alpha \in \omega_1$. A symmetrical argument shows that there is a further $r \leq q$ that forces $\check{f}(\dot{h}(\check{\beta})) = \check{j}'$, so that $\check{f}$ is forced by r not to be constant on $\mathrm{im}(\dot{h})$.

Since $g^{-1}[\{i\}] \cap E^{\omega_2}_\omega \cap f^{-1}[\{j\}]$ is stationary in ω_2, we can find $M \prec H(\Theta)$ countable containing $\mathbb{P}, p$ and $\dot{h}$ such that $\sup(M \cap \omega_2) \in g^{-1}[\{i\}] \cap E^{\omega_2}_\omega \cap f^{-1}[\{j\}]$ (as in the proof of Lemma 1.9). Let $(D_n)_{n\in\omega}$ enumerate all open dense subsets of $\mathbb{P}$ lying in M. Find a descending sequence $(p_n)_{n\in\omega}$ with $p_0 \leq p$ such that $p_n \in D_n$ for all $n \in \omega$. Define $\delta_1 := M \cap \omega_1$ and $\delta_2 := \sup(M \cap \omega_2)$ and let $q := \bigcup_n p_n \cup \{(\delta_1, \delta_2)\}$.

Subclaim. *q is a condition in $\mathbb{P}$.*

Proof. Let $q' := \bigcup_n p_n$. It suffices to show that $\mathrm{dom}(q') = M \cap \omega_1$ and $\sup(\mathrm{im}(q')) = \sup(M \cap \omega_2)$. For any $n \in \omega$, $p_n \in M$, so $p_n \subseteq M$ as it is countable, which implies $q' \subseteq M$ and thus $\mathrm{dom}(q') \subseteq M \cap \omega_1$ and $\mathrm{im}(q') \subseteq M \cap \omega_2$. On the other hand, for any $\alpha < M \cap \omega_1$, the set of all conditions r with $\alpha \in \mathrm{dom}(r)$ is open dense in $\mathbb{P}$ and in M, so $\alpha \in \mathrm{dom}(q')$. Additionally, for any $\alpha < \sup(M \cap \omega_2)$, there is $\beta > \alpha$ with $\beta \in M \cap \omega_2$. The set of all conditions r with $\beta < \sup(\mathrm{im}(r))$ is open dense in $\mathbb{P}$ and in M, so $\beta < \sup(\mathrm{im}(q'))$. □

A similar argument shows the following:

Subclaim. *q forces $\dot{h}(\check{\delta}_1) = \check{\delta}_2$.*

Proof. Because $\dot{h}$ is forced to be continuous, it suffices to show that q forces $\sup(\dot{h}[\check{\delta}_1]) = \check{\delta}_2$. Whenever $\alpha \in \delta_1$, the set D_α of conditions r which decide $\dot{h}(\check{\alpha})$ is open dense in $\mathbb{P}$ and in M, so there is $n \in \omega$ with $p_n \in D_\alpha$. The corresponding value that is decided for $\dot{h}(\check{\alpha})$ is in M as well by elementarity. This shows $\sup(\dot{h}[\check{\delta}_1]) \leq \check{\delta}_2$. On the other hand, given any $\beta \in M \cap \omega_2$ the set E_β of conditions r which force $\sup(\mathrm{im}(\dot{h})) \geq \check{\beta}$ is open dense in $\mathbb{P}$ and in M, so again there is $n \in \omega$ with $p_n \in E_\beta$. This shows the equality. □

So in particular q forces that $\check{\delta}_2 \in \mathrm{im}(\dot{h})$. Because $\delta_2 \in f^{-1}[\{j\}]$, q forces that $\check{f}(\dot{h}(\check{\alpha})) = \check{j}$ for some $\alpha \in \omega_1$. □

Now let, for $\alpha < \omega_1$, $D_\alpha := \{r \in \mathbb{P} \mid \alpha \in \mathrm{dom}(r)\}$. It follows that any D_α is open dense in $\mathbb{P}$ (using similar arguments to before). Thus, by $\mathrm{MM}_{F(\lambda)}(f)$ there is a filter $G \subseteq \mathbb{P}$ which intersects any D_α. It follows that $\bigcup G$ is a normal function from ω_1 into ω_2 such that g is constant (with value i) on $\mathrm{im}(\bigcup G)$. □

As a corollary, we obtain that there is no implication from $F(\lambda', \omega_2)$ to $F(\lambda, \omega_2)$ when $\lambda' < \lambda$, i.e. it is consistent that there is a partition of ω_2 into 17 parts such that none of them contain a closed copy of ω_1 while there is no such partition of ω_2 into any fewer parts.

Let us note the following two results of a similar type: In [2], Theorem 8.1, Cummings, Foreman and Magidor show that for any $\eta \leq \omega_1$ it is consistent that there are η stationary subsets of $\omega_2 \cap \mathrm{cof}(\omega)$ which do not reflect simultaneously while every collection of fewer than η stationary subsets of $\omega_2 \cap \mathrm{cof}(\omega)$ does reflect simultaneously. In [3], the same authors show that there is no implication from $\square_{\omega_1,\lambda}$ to $\square_{\omega_1,\lambda'}$ for $\lambda' < \lambda$.

However, there are technical differences between the methods in the previous papers and the ones in this paper: In the previous papers, a two-step iteration $\mathbb{P} * \dot{\mathbb{Q}}$ was used such that $\mathbb{P}$ adds a "bad" set and $\dot{\mathbb{Q}}$ then "destroys" it, e.g. $\mathbb{P}$ adds a nonreflecting stationary set and $\dot{\mathbb{Q}}$ adds a club set disjoint from it. The two-step iteration itself then is well-behaved (in most cases, it is equivalent to Cohen forcing) and thus preserves various large cardinal properties. It is then shown that certain properties are true even in the intermediate extension by $\mathbb{P}$ (this method was used for example by Hayut-Lambie-Hanson in [11]) On the other hand, in this paper, we do not have to use the second iterand $\dot{\mathbb{Q}}$ since our original poset already suffices for the preservation of large cardinal properties because of its $<\kappa$-$<\omega_2$-directed closure.

Corollary 4.2. *Assume κ is a supercompact cardinal and $\lambda \in [2, \omega_1] \cup \{\kappa\}$. There is a forcing extension in which $\kappa = \omega_2$, no cardinals above κ are collapsed, $F(\lambda, \kappa)$ fails and $F(\lambda', \kappa)$ holds for all $\lambda' < \lambda$.*

Proof. By Theorem 3.7, there is a forcing extension where $\mathrm{MM}_{F(\lambda)}(f)$ holds for some function $f\colon \omega_2 \to \lambda$. This function witnesses $\neg F(\lambda, \kappa)$. On the other hand, Theorem 4.1 implies that $F(\lambda', \kappa)$ holds for all $\lambda' < \lambda$. □

5. Separating Instances of F^+

As Fuchs showed in [8], there is a natural implication between instances of SRP and instances of F^+: As we defined in the introduction, given a partition $(D_i)_{i\in\omega_1}$ of ω_1 and a sequence $(A_i)_{i\in\omega_1}$ of stationary subsets of E^κ_ω, there is for any $\Theta \geq \kappa$ a canonical subset of $[H(\Theta)]^{<\omega_1}$ associated with the pair $((D_i)_{i\in\omega_1}, (A_i)_{i\in\omega_1})$:

$$S((D_i)_{i\in\omega_1},(A_i)_{i\in\omega_1},\Theta):=$$
$$\{M\in[H(\Theta)]^{<\omega_1}\mid\forall i\in\omega_1(M\cap\omega_1\in D_i\to\sup(M\cap\kappa)\in A_i)\}$$

We also define $\mathcal{S}((D_i)_{i\in\omega_1},\kappa,\Theta)$ as the collection of $S((D_i)_{i\in\omega_1},(A_i)_{i\in\omega_1},\Theta)\cap C$ for all sequences $(A_i)_{i\in\omega_1}$ of stationary subsets of E^κ_ω and clubs $C\subseteq[H(\Theta)]^{<\omega_1}$.

We first answer a question from Fuchs' paper by showing that this set is projective stationary for any partition of ω_1:

Theorem 5.1. *Let $(D_i)_{i\in\omega_1}$ be a partition of ω_1, $\kappa\geq\omega_2$ regular and $(A_i)_{i\in\omega_1}$ a sequence of stationary subsets of E^κ_ω. Then $S((D_i)_{i\in\omega_1},(A_i)_{i\in\omega_1},\Theta)$ is projective stationary.*

Proof. Let $C\subseteq[H(\Theta)]^{<\omega_1}$ be club and $S\subseteq\omega_1$ stationary. Our aim is to find $M\in C\cap S((D_i)_{i\in\omega_1},(A_i)_{i\in\omega_1},\Theta)$ with $M\cap\omega_1\in S$. Let $\mathcal{H}$ be an expansion of $H(\Theta)$ coding all necessary information such that any $M\prec\mathcal{H}$ is in C.

Let T be the tree $\kappa^{<\omega}$, i.e. T consists of all finite sequences of elements of κ. Define a function $f\colon[T]\to\omega_1$ by letting $f(b)$ be the minimum $\alpha\in S$ such that $\mathrm{Hull}^{\mathcal{H}}(b\cup\alpha)\cap\omega_1=\alpha$. Such an α exists as the set of all β with $\mathrm{Hull}^{\mathcal{H}}(b\cup\beta)\cap\omega_1=\beta$ is club in ω_1.

It follows that for any $\alpha<\omega_1$, the set $f^{-1}[\alpha+1]$ is closed in the tree topology: If $f(b)>\alpha$ then there is a finite $\eta\sqsubset b$ such that $\mathrm{Hull}^{\mathcal{H}}(\eta\cup\alpha)\cap\omega_1>\alpha$. Consequently, $\{c\in\kappa^{<\omega}\mid\eta\sqsubset c\}$ is open, contains b and is disjoint from $f^{-1}[\alpha+1]$.

By Lemma 3.5 in [19, chapter XI], there exists a tree $T''\leq T$ and $\beta\in\omega_1$ such that $|\mathrm{Suc}_{T''}(\eta)|=\kappa$ for any $\eta\in T''$ and $[T'']\subseteq f^{-1}[\beta+1]$. Let $\alpha\in\omega_1$ be minimal such that for some $\eta\in T''$ and any branch b of T'' containing η, $f(b)\leq\alpha$. Let

$$T':=T''\restriction\eta:=\{\eta'\in T''\mid\eta'\sqsubseteq\eta\vee\eta\sqsubseteq\eta'\}$$

It follows from the minimality of α that for any $\eta\in T'$ there is a branch b of T' with $f(b)=\alpha$: We know $f(b)\leq\alpha$ and if $f(b)<\alpha$ for all such branches, η would witness that α was not minimal.

However, this actually implies that $\mathrm{Hull}^{\mathcal{H}}(b\cup\alpha)\cap\omega_1=\alpha$ for any branch b of T': Otherwise there would be a finite $\eta\sqsubset b$ such that $\mathrm{Hull}^{\mathcal{H}}(\eta\cup\alpha)\cap\omega_1>\alpha$ but then we could never extend η to a branch b' with $f(b')=\alpha$.

Now consider the following clubs:

(1) Let $C_0\subseteq\kappa$ consist of all those $\gamma\in\kappa$ such that $\sup(\mathrm{Hull}^{\mathcal{H}}(\alpha\cup s)\cap\kappa)<\gamma$ for all $s\in[\gamma]^{<\omega}$.
(2) Let $C_1\subseteq\kappa$ consist of all those $\gamma\in\kappa$ such that whenever $\eta\in T'$ is above the stem with $\eta\in\gamma^{<\omega}$ and $\alpha\in\gamma$ there is $\beta\in(\alpha,\gamma)$ such that $\eta^\frown\beta\in T'$.

C_0 is clearly club in κ. C_1 is club in κ as well because whenever M is an elementary submodel of $(H(\Theta),T',\in)$ has size ω_1, $M\cap\kappa$ is in C_1 (this shows unboundedness, closure is clear).

Let i be such that $\alpha\in D_i$. Let $\gamma\in A_i\cap C_0\cap C_1$. Fix sequences $(\alpha_n)_{n\in\omega}$ converging to α and $(\gamma_n)_{n\in\omega}$ converging to γ (this is possible as $A_\alpha\subseteq\mathrm{cof}(\omega)$).

We construct a branch b of T' as follows: Let η_0 be the stem of T'. Given η_n, find η_{n+1} in T' such that $\max(\eta_{n+1})>\gamma_n$. Let $b:=\bigcup_n\eta_n$ and let $M:=\mathrm{Hull}^{\mathcal{H}}(\alpha\cup b)\in C$. By assumption we have $M\cap\omega_1=\alpha\in S\cap D_i$. Furthermore we have $\gamma\leq\sup(M\cap\kappa)$ by the construction of b and $\gamma\geq\sup(M\cap\kappa)$ by $\gamma\in C_0$, so $\sup(M\cap\kappa)\in A_i$.

Consequently, $M\in C\cap S((D_i)_{i\in\omega_1},(A_i)_{i\in\omega_1},\Theta)$ and $M\cap\omega_1\in S$. □

It is clear that any intersection of a projective stationary set with a club is projective stationary. Following an argument of Fuchs, this shows that SRP implies $F^+((D_i)_{i\in\omega_1},\kappa)$ for any partition $(D_i)_{i\in\omega_1}$ of ω_1:

Theorem 5.2. *Let $(D_i)_{i\in\omega_1}$ be a partition of ω_1 and $\kappa \geq \omega_2$ regular. Let $\Theta \geq \kappa$ be sufficiently large. Then $\mathrm{SRP}(\mathcal{S}((D_i)_{i\in\omega_1}, \kappa, \Theta))$ implies $F^+((D_i)_{i\in\omega_1}, \kappa)$.*

Proof. Let $(A_i)_{i\in\omega_1}$ be a sequence of stationary subsets of E^κ_ω. Let $\Theta \geq \kappa$ be large enough. Let $C \subseteq [H(\Theta)]^{<\omega_1}$ be the club of all $M \prec H(\Theta)$ such that $(D_i)_{i\in\omega_1}, (A_i)_{i\in\omega_1} \in M$. Since $\mathrm{SRP}(\mathcal{S}((D_i)_{i\in\omega_1}, \kappa, \Theta))$ holds, there is a sequence $(X_i)_{i\in\omega_1}$ of elements of $S((D_i)_{i\in\omega_1}, (A_i)_{i\in\omega_1}, \Theta) \cap C$ such that $X_i \in X_{i+1}$ for any $i \in \omega_1$ and $X_i = \bigcup_{j<i} X_j$ for any limit $j \in \omega_1$. Let $D \subseteq \omega_1$ be the club consisting of $X_i \cap \omega_1$ for $i \in \omega_1$ and define f on D by letting $f(X_i \cap \omega_1) := \sup(X_i \cap \kappa)$. Then f is a normal function from D into ω_2 and satisfies $f[D_i \cap D] \subseteq A_i$. So all that is left is to "fill in the gaps".

For $i \in \omega_1$, let $\mathbb{P}_i$ be the forcing notion of all functions p with $\mathrm{dom}(p) = [X_i \cap \omega_1, \alpha]$ where $\alpha < \omega_1$ such that $p(\beta) \in A_j$ whenever $\beta \in D_j \cap \mathrm{dom}(p)$ for all $j \in \omega_1$. Because $X_i \in X_{i+1}$, $\mathbb{P}_i \in X_{i+1}$. Let $(D^i_n)_{n\in\omega}$ enumerate all open dense subsets of $\mathbb{P}_i$ lying in X_{i+1}. Let $(p^i_n)_{n\in\omega}$ be a descending sequence of elements of $\mathbb{P}_i \cap X_{i+1}$ such that $p^i_n \in D^i_n$ for any $n \in \omega$. Let $p_i := \bigcup_n p^i_n$. It follows that $\mathrm{dom}(p_i) = [X_i \cap \omega_1, X_{i+1} \cap \omega_1)$ and $\sup(\mathrm{im}(p)) = \sup(X_{i+1} \cap \kappa)$ by genericity. Ergo the function g defined by

$$g := f \cup \bigcup_{i<\omega_1} p_i$$

is a normal function from ω_1 into κ such that $g[D_i] \subseteq A_i$ for any $i \in \omega_1$. □

We now investigate implications between instances of F^+. We first introduce two partial orders on the set of partitions of ω_1:

Definition 5.3. Let $\overline{D} = (D_i)_{i\in\omega_1}$ and $\overline{E} = (E_i)_{i\in\omega_1}$ be partitions of ω_1.

(1) We write $\overline{D} \leq \overline{E}$ if there is a function $h\colon \omega_1 \to \omega_1$ such that for any $i \in \omega_1$, $D_i \subseteq E_{h(i)}$.
(2) We write $\overline{D} \leq^* \overline{E}$ if there is a function $h\colon \omega_1 \to \omega_1$ and a club $C \subseteq \omega_1$ such that for any $i \in \omega_1$, $D_i \cap C \subseteq E_{h(i)}$.

So $\overline{D} \leq \overline{E}$ states that $\overline{D}$ refines $\overline{E}$ while $\overline{D} \leq^* \overline{E}$ states that $\overline{D}$ refines $\overline{E}$ on a club.

Remark 5.4. The following reformulation of $\leq^*$ is useful: $\overline{D} \leq^* \overline{E}$ if and only if there is a function $h\colon \omega_1 \to \omega_1$ such that the set

$$S(h) := \{\alpha \in \omega_1 \mid \exists i(\alpha \in D_i \smallsetminus E_{h(i)})\}$$

is nonstationary: Assume $\overline{D} \leq^* \overline{E}$, witnessed by h and C. Then for any $\alpha \in C$ and any $i \in \omega_1$, if $\alpha \in C \cap D_i$, $\alpha \in E_{h(i)}$. Thus C is disjoint from $S(h)$. On the other hand, assume there is h such that $S(h)$ is nonstationary, with $C \subseteq \omega_1$ club and disjoint from $S(h)$. Then for any $\alpha \in C$ and all $i \in \omega_1$, $\alpha \in D_i$ implies $\alpha \in E_{h(i)}$, so $D_i \cap C \subseteq E_{h(i)} \cap C$.

This enables us to show the following:

Lemma 5.5. *Assume $(D_i)_{i\in\omega_1}$ and $(E_i)_{i\in\omega_1}$ are partitions of ω_1 such that $\overline{D} \leq \overline{E}$ and $\kappa \geq \omega_2$ is regular. Then $F^+(\overline{D}, \kappa)$ implies $F^+(\overline{E}, \kappa)$.*

Proof. Let h witness $\overline{D} \leq \overline{E}$.

Let $(B_i)_{i\in\omega_1}$ be a sequence of stationary subsets of E^κ_ω. Define $(A_i)_{i\in\omega_1}$ by letting $A_i := B_{h(i)}$. Applying $F^+(\overline{D}, \kappa)$, let $f\colon \omega_1 \to \kappa$ be a normal function such that $f[D_i] \subseteq A_i$. Let $j \in \omega_1$ and $\alpha \in E_j$. Let $i \in \omega_1$ be such that $\alpha \in D_i$. Then $h(i) = j$, since $D_i \subseteq E_{h(i)}$ and $(D_i)_{i\in\omega_1}$, $(E_i)_{i\in\omega_1}$ are partitions. Ergo

$$f(\alpha) \in A_i = B_{h(i)} = B_j$$

In summary, $f[E_j] \subseteq B_j$ for any $j \in \omega_1$. □

It turns out that, with more work, we can actually do better and show that the weaker partial order $\leq^*$ suffices to guarantee an implication between the corresponding instances of F^+. We start with a modification of Friedman's original result in [7] regarding closed sets of ordinals:

Lemma 5.6. *Suppose $\kappa \geq \omega_1$ is a regular cardinal and $(A_i)_{i\in\omega_1}$ is a sequence of stationary subsets of E^κ_ω. For each $\sigma < \omega_1$, there is a normal function $f\colon \sigma \to \kappa$ such that $f(i) \in A_i$ for every $i < \sigma$.*

Proof. We prove a stronger statement by induction on σ, following Friedman's original proof. Let $(A_i)_{i\in\omega_1}$ be a sequence of stationary subsets of E^κ_ω.

Claim. *For every $\sigma < \omega_1$ and $\gamma < \kappa$ there is a normal function $f\colon \sigma \to \kappa$ such that $f(0) > \gamma$ and $f(i) \in A_i$ for every $i < \sigma$.*

Proof. By induction on σ for every γ simultaneously. Suppose the statement holds for all $\beta < \sigma$.

(1) Case 1: $\sigma = \beta + 2$ for some β. Let γ be arbitrary. By the inductive hypothesis, let $f\colon \beta + 1 \to \kappa$ be a normal function with $f(0) > \gamma$ and $f(i) \in A_i$ for every $i < \beta + 1$. Then simply let $f(\beta + 1)$ be an arbitrary element of $A_{\beta+1}$ above $f(\beta)$. Clearly, f is as required.

(2) Case 2: σ is a limit. Let γ be arbitrary and write $\sigma = \bigcup_{n\in\omega} \beta_n$, where $\beta_0 = 0$ and every β_n, $n > 0$ is a successor ordinal. By the inductive hypothesis, let f be as follows: $f \restriction [\beta_n, \beta_{n+1})\colon [\beta_n, \beta_{n+1}) \to \kappa$ is normal such that $f(0) > \gamma$, $f(\beta_n) > f(\beta_n - 1)$ and $f_n(i) \in A_i$ for every $i < \sigma$. Since we have obtained f by stitching together normal functions at successor ordinals (and $f(\beta_n) > f(\beta_n - 1)$), f itself is a normal function. Additionally, $f(i) \in A_i$ for every $i < \sigma$, so it is as required.

(3) Case 3: $\sigma = \beta + 1$ for a limit β. Write $\beta = \bigcup_{n\in\omega} \beta_n$, $\beta_0 = 0$ and every β_n, $n > 0$ is a successor ordinal. Using the inductive hypothesis, define a sequence of normal functions $(f_\xi)_{\xi<\kappa}$ from β into κ such that
   (a) $f_0(0) > \gamma$.
   (b) For every $\xi < \kappa$ and $i < \beta$, $f_\xi(i) \in A_i$.
   (c) If $\xi_0 < \xi_1$ then $f_{\xi_1}(0) > \sup(\mathrm{im}(f_{\xi_0}))$.

   Define $g\colon \kappa \to \kappa$ by $g(\xi) := \sup_{\zeta<\xi} \sup(\mathrm{im}(f_\zeta))$. Then g is a normal function and therefore its image is club in κ. Let $\xi < \kappa$ be a limit ordinal such that $g(\xi) \in A_\beta$. Since A_β consists of ordinals of countable cofinality, $\mathrm{cf}(g(\xi)) = \omega$ which equals $\mathrm{cf}(\xi)$ by the normality of g. So let $\xi = \bigcup_{n\in\omega} \xi_n$, the ξ_n ascending. Define $f\colon \sigma \to \kappa$ as follows: For every n, $f \restriction [\beta_n, \beta_{n+1}) = f_{\xi_n} \restriction [\beta_n, \beta_{n+1})$. Finally, let $f(\beta) = g(\xi)$.

   Again, f is normal, since we have stitched together normal functions at successor ordinals and $f(\beta_n) = f_{\xi_n}(\beta_n) > f_{\xi_{n-1}}(\beta_n - 1)$. Additionally, $g(\xi) = \bigcup_{n\in\omega} g(\xi_n)$ by normality and, for every $n \in \omega$, $f(\beta_n) = f_{\xi_n}(\beta_n) > g(\xi_{n-1})$, so $\sup(\mathrm{im}(f \restriction \beta)) = g(\xi)$. Therefore, f is normal. Lastly, for every $i < \beta$, $f(i) \in A_i$ by construction and $f(\beta) = g(\xi) \in A_\beta$ by assumption, so f is as required.

□

This finishes the proof. □

In particular, we have the following:

Corollary 5.7. *Suppose $\kappa \geq \omega_1$ is a regular cardinal, $(D_i)_{i\in\omega_1}$ is a partition of ω_1 and $(A_i)_{i\in\omega_1}$ is a sequence of stationary subsets of E^κ_ω. For every $\sigma < \omega_1$, there is a normal function $f\colon \sigma \to \kappa$ such that $f[D_i] \subseteq A_i$ for every $i < \sigma$.*

Proof. Given a sequence $(A_i)_{i\in\omega_1}$ as in the lemma, let $(B_i)_{i\in\omega_1}$ be defined such that $B_i = A_j$ where $i \in D_j$. Then let $f\colon \sigma \to \kappa$ be such that $f(i) \in B_i$ for every $i \in \sigma$. Let $i < \sigma$ and let j be such that $i \in D_j$. Then $B_i = A_j$ and so $f(i) \in A_j$. It follows that $f[D_j] \subseteq A_j$ for every $j \in \omega_1$. □

This shows that, roughly speaking, the countable version of $F^+((D_i)_{i\in\omega_1}, \kappa)$ holds for every partition $(D_i)_{i\in\omega_1}$ and every regular cardinal κ on the basis of ZFC. It also allows us to show that $\leq^*$ characterizes the implication between instances of F^+:

Lemma 5.8. *Assume $(D_i)_{i\in\omega_1}$ and $(E_i)_{i\in\omega_1}$ are partitions of ω_1 such that $\overline{D} \leq^* \overline{E}$ and $\kappa \geq \omega_2$ is regular. Then $F^+(\overline{D}, \kappa)$ implies $F^+(\overline{E}, \kappa)$.*

Proof. Let h and C witness $\overline{D} \leq^* \overline{E}$, i.e. $h\colon \omega_1 \to \omega_1$ is such that for every $i \in \omega_1$, $D_i \cap C \subseteq E_{h(i)}$.

Let $(B_i)_{i\in\omega_1}$ be a sequence of stationary subsets of E^κ_ω. Define $(A^*_i)_{i\in\omega_1}$ by letting $A^*_i := B_{h(i)}$. Let $E \subseteq \kappa$ be the set of all $\delta \in \kappa$ such that whenever $\gamma < \delta$ and $\sigma < \omega_1$, there is a normal function $f\colon \sigma \to \kappa$ such that $f[E_i \cap \sigma] \subseteq B_i$ for every $i \in \omega_1$ and $f[\sigma] \subseteq (\gamma, \delta)$. By Corollary 5.7 and standard arguments, E is club in κ and so $A_i := A^*_i \cap E$ is stationary.

Now apply $F^+(\overline{D}, \kappa)$ to find $f\colon \omega_1 \to \kappa$ normal such that $f[D_i] \subseteq A_i$. Let $j \in \omega_1$ and $\alpha \in E_j \cap C$. Let $i \in \omega_1$ be such that $\alpha \in D_i$. Then $h(i) = j$, since $D_i \cap C \subseteq E_j$ and $\overline{D}$, $\overline{E}$ are partitions. Ergo,

$$f(\alpha) \in A_i \subseteq A^*_i = B_{h(i)} = B_j$$

So, for every $j \in \omega_1$, $f[E_j \cap C] \subseteq B_j$ and we only have to modify f on a nonstationary set. So let $h\colon \omega_1 \to C$ be a normal function. For simplicity, we assume that C only contains limit ordinals. By induction on $i \in \omega_1$, we define a normal function $f^* \restriction h(i) + 1$ such that $f^*(h(j)) = f(h(j))$ and $f^*[E_j] \subseteq B_j$ for every $j \in \omega_1$.

First assume that i is a limit and we have defined $f^* \restriction h(j) + 1$ for every $j < i$. Simply let $f^*(h(i)) = f(h(i))$. Then f^* is normal since it is an ascending union of normal functions and $f^*(h(i)) = f(h(i)) = \sup_{j<i} f(h(j)) = \sup_{j<i} f^*(h(j)) = \sup \operatorname{im}(f^* \restriction h(i))$ since f and h are normal. Additionally, $f^*[E_j] \subseteq B_j$ for every $j \in \omega_1$ since f satisfies this.

Now assume that $i = j + 1$ and we have defined $f^* \restriction h(j) + 1$. We know that $f(h(i)) > f(h(j))$ and $f(h(i)) \in E$. Ergo for every $\sigma < \omega_1$ and $\gamma < f(h(i))$ there is a normal function $f\colon \sigma \to \kappa$ such that $f[E_i] \subseteq B_i$ for every $i < \sigma$ and $f[\sigma] \subseteq (\gamma, h(i+1))$. Since $f(h(i)) > f(h(j))$ has cofinality ω, write $f(h(i)) = \bigcup_{n\in\omega} \zeta_n$, where $\zeta_0 = f(h(j))$. We know that $h(i+1)$ is a limit, so write $h(i+1) = \bigcup_{n\in\omega} \beta_n$, where $\beta_0 = h(i)+1$ and β_n for $n > 0$ is a successor. Define f^* such that $f^* \restriction [\beta_n, \beta_{n+1})$ is any normal function with $f^*[E_i] \subseteq A^*_i$ for every $n \in \omega$ and $f^*(\beta_{n+1}) > \zeta_n$. Lastly, let $f^*(h(i)) := f(h(i))$.

Since f^* is comprised of normal functions glued along successor ordinals, $f^* \restriction h(i)$ is normal itself. Furthermore, by construction, $f^* \restriction h(i)$ is unbounded in $f(h(i))$, so $f^* \restriction h(i) + 1$ is normal. Lastly, by construction, $f^*[E_j] \subseteq B_j$ for every $j \in \omega_1$. This finishes the construction. □

Whenever $(D_i)_{i\in\omega_1}$ is a partition of ω_1 where one D_i is club in ω_1 and $(E_i)_{i\in\omega_1}$ is a partition with $E_0 = \omega_1$, $(E_i)_{i\in\omega_1} \leq^* (D_i)_{i\in\omega_1}$: Let i be such that D_i is club in ω_1. Then h defined by $h(0) = i$, $h(j)$ arbitrary otherwise and D_i witness that $(E_i)_{i\in\omega_1}$ refines $(D_i)_{i\in\omega_1}$ on a club. In particular, $F^+((D_i)_{i\in\omega_1}, \kappa)$ is equivalent to $F^+((E_i)_{i\in\omega_1}, \kappa)$ and thus $F^+((D_i)_{i\in\omega_1}, \omega_2)$ has consistency strength exactly a Mahlo cardinal.

On the other hand, if at least two D_i are stationary in ω_1, $F^+((D_i)_{i\in\omega_1}, \kappa)$ implies that any two stationary subsets of E^κ_ω reflect simultaneously. In particular,

$F^+((D_i)_{i\in\omega_1},\omega_2)$ implies that ω_2 is weakly compact in L. Again, this is the exact consistency strength by results of Shelah.

We now show that $\leq^*$ exactly characterizes the implication between instances of F^+. Even stronger, we obtain that in every model of $\mathrm{MM}_{F^+((D_i)_{i\in\omega_1})}((A_i)_{i\in\omega_1})$, we do not just have $F^+((E_i)_{i\in\omega_1})$ for every partition $(E_i)_{i\in\omega_1}$ which is not $\leq^*$-below $(D_i)_{i\in\omega_1}$, we actually have that the corresponding fragment of SRP holds.

Theorem 5.9. *Assume $(D_i)_{i\in\omega_1}$ is a partition of ω_1 and $(A_i)_{i\in\omega_1}$ is a disjoint sequence of stationary subsets of $E^{\omega_2}_\omega$ such that $\mathrm{MM}_{F^+((D_i)_{i\in\omega_1})}((A_i)_{i\in\omega_1})$ holds. Let $(E_i)_{i\in\omega_1}$ be a partition of ω_1. Then the following are equivalent:*

(1) $\overline{E} \not\leq^* \overline{D}$
(2) $\mathrm{SRP}(\mathcal{S}((E_i)_{i\in\omega_1},\omega_2,\Theta))$ *holds for all $\Theta \geq \kappa$.*

Proof. If $\mathrm{SRP}(\mathcal{S}((E_i)_{i\in\omega_1},\omega_2,\Theta))$ even just holds for $\Theta=\kappa$, then $F^+((E_i)_{i\in\omega_1},\kappa)$ holds by Theorem 5.2. Since $F^+((D_i)_{i\in\omega_1})$ fails by assumption, Lemma 5.8 shows that $\overline{E} \not\leq^* \overline{D}$.

So let $(B_i)_{i\in\omega_1}$ be a sequence of stationary subsets of $E^{\omega_2}_\omega$ and $C \subseteq [H(\Theta)]^{<\omega_1}$ any club. Let $\mathbb{P}$ be the forcing shooting an $\in$-increasing and continuous sequence through $S := S((E_i)_{i\in\omega_1},(B_i)_{i\in\omega_1},\Theta)\cap C$, i.e. $\mathbb{P}$ consists of continuous $\in$-increasing functions $p\colon \alpha+1 \to S$, where $\alpha<\omega_1$. By the arguments in [4, Theorem 1.1], since S is projective stationary by Theorem 5.1, $\mathbb{P}$ preserves stationary subsets of ω_1. We now aim to show that $\mathbb{P}$ does not add a normal function $g\colon \omega_1\to\omega_2$ with $g[D_i]\subseteq A_i$ for all $i\in\omega_1$. Assume otherwise and let $\dot{g}$ be a name for such a function.

Claim. *For any $i\in\omega_1$ there exists at most one $h(i)\in\omega_1$ such that $B_i \smallsetminus A_{h(i)}$ is nonstationary.*

Proof. Assume otherwise that $B_i\smallsetminus A_j$ and $B_i\smallsetminus A_{j'}$ are nonstationary for $j\neq j'$. Then $B_i = B_i\smallsetminus(A_j\cap A_{j'}) = (B_i\smallsetminus A_j)\cup(B_i\smallsetminus A_{j'})$ is nonstationary as well, a contradiction. □

Consider the function h defined by using the claim (and letting $h(i):=0$ if such a value does not exist). Then for all $i,j\in\omega_1$, if $j\neq h(i)$, $B_i\smallsetminus A_j$ is stationary. Because $\overline{E}\not\leq^*\overline{D}$, the set

$$S' := \{\alpha\in\omega_1 \mid \exists i(\alpha\in E_i\smallsetminus D_{h(i)})\}$$

is stationary (see Remark 5.4).

Now we proceed as in the proof of Theorem 5.1: Let $\mathcal{H}$ be an algebra on $H(\Theta)$ such that any $M\in[H(\Theta)]^{<\omega_1}$ with $M\prec\mathcal{H}$ is in C. Find a Laver-style tree $T\leq(\omega_2)^{<\omega}$ (possibly with a stem) and $\alpha\in S'$ such that for any branch b of T, $\mathrm{Hull}^{\mathcal{H}}(b\cup\alpha)\cap\omega_1=\alpha$. Let $i\in\omega_1$ be such that $\alpha\in E_i\smallsetminus D_{h(i)}$ and let j be such that $\alpha\in D_j$. Then $j\neq h(i)$, so $B_i\smallsetminus A_j$ is stationary. Ergo we can find $\gamma\in B_i\smallsetminus A_j$ which is a "limit point of the tree T" (as in the proof of Theorem 5.1). Thus there is $M\prec\mathcal{H}$ such that $M\cap\omega_1\in E_i\cap D_j$ and $\sup(M\cap\omega_2)\in B_i\smallsetminus A_j$. Let $(D_n)_{n\in\omega}$ enumerate all open dense subsets of $\mathbb{P}$ lying in M, let $(p_n)_{n\in\omega}$ be a descending sequence of elements of $\mathbb{P}$ with $p_n\in D_n$ and let $p:=\bigcup_n p_n\cup\{(M\cap\omega_1,M)\}$. It follows that $p\in\mathbb{P}$ (since $M\in S$ and $M=\bigcup_n p_n(\mathrm{dom}(p_n)-1)$) and p forces $\dot{g}(M\cap\omega_1)=\sup(M\cap\omega_2)$, so p forces $\dot{g}[\check{D}_j]\not\subseteq\check{A}_j$, a contradiction.

Now let $(D_i)_{i\in\omega_1}$ be defined by $D_i:=\{p\in\mathbb{P}\mid i\in\mathrm{dom}(p)\}$. It follows that there exists a filter G on $\mathbb{P}$ intersecting every D_i and that the union of this filter is as required. □

Fuchs asked in [8] whether there is an implication from $\mathrm{SRP}(\mathcal{S}((D_i)_{i\in\omega_1},\kappa,\Theta))$ to $F^+((E_i)_{i\in\omega_1},\omega_2)$, where $(D_i)_{i\in\omega_1}$ is the trivial partition given by $D_0:=\omega_1$ and $D_i:=\emptyset$ for $i\neq 0$ and $(E_i)_{i\in\omega_1}$ is a maximal partition of ω_1 into stationary sets,

i.e. every E_i is stationary and for every stationary $A \subseteq \omega_1$ there is $i \in \omega_1$ such that $A \cap E_i$ is stationary. We can answer this question in the negative in a strong way by showing the following:

Lemma 5.10. *Let $(D_i)_{i\in\omega_1}$ be the trivial partition of ω_1 and let $(E_i)_{i\in\omega_1}$ be a partition of ω_1 into at least two stationary sets. Then $\overline{D} \not\leq^* \overline{E}$.*

Proof. Let $h\colon \omega_1 \to \omega_1$ be any function. Let $i := h(0)$. So there is a club C such that $D_0 \cap C \subseteq E_{h(0)}$. But this is a contradiction as $E_{h(0)}$ has a stationary complement. □

Thus we have the following:

Corollary 5.11. *Assume we are in the standard model of Martin's Maximum. Assume $(D_i)_{i\in\omega_1}$ is the trivial partition of ω_1 and $(E_i)_{i\in\omega_1}$ is a partition of ω_1 such that at least two $E_i, E_{i'}$ are stationary. Then there is a forcing extension where $\mathrm{SRP}(\mathcal{S}((D_i)_{i\in\omega_1}, \omega_2, \Theta))$ holds and $F^+((E_i)_{i\in\omega_1}, \omega_2)$ fails.*

Proof. Simply force with $\mathbb{P}_{F^+((E_i)_{i\in\omega_1})}(\omega_2)$ to obtain $V[G]$. In the resulting model, let $F := \bigcup G$ and $A_i := \{\alpha \in \omega_2 \mid F(\alpha, i) = 1\}$. Then $\mathrm{MM}_{F^+((E_i)_{i\in\omega_1})}((A_i)_{i\in\omega_1})$ holds (so $F^+((E_i)_{i\in\omega_1}, \omega_2)$ fails), and $(A_i)_{i\in\omega_1}$ is a disjoint sequence of stationary subsets of $E^{\omega_2}_\omega$. By Theorem 5.9 and Lemma 5.10, $\mathrm{SRP}(\mathcal{S}((D_i)_{i\in\omega_1}, \omega_2, \Theta))$ holds. □

More subtle separations of instances of F^+ and SRP are also possible: E.g. it is consistent that there are two maximal partitions $(E_i)_{i\in\omega_1}$ and $(D_i)_{i\in\omega_1}$ of ω_1 into stationary sets such that $\mathrm{SRP}(\mathcal{S}((D_i)_{i\in\omega_1}, \omega_2, \Theta))$ holds for all $\Theta \geq \omega_2$ and $F^+((E_i)_{i\in\omega_1}, \omega_2)$ fails: Assume we are in the standard model of MM. Let $(F_i)_{i\in\omega_1}$ be a maximal partition of ω_1 into stationary sets which can be constructed as follows (see [8], Remark 3.17): Let $(G_i)_{i\in\omega_1}$ be any partition of ω_1 such that each G_i is stationary. Let $f\colon \omega_1 \to \omega_1$ be defined as follows: If $j \in G_i$ for $i < j$, let $f(j) := i$, otherwise let $f(j) := 0$. Ergo f is regressive. For any $i \in \omega_1$, let $F_i := f^{-1}[\{i\}]$. As $F_i \supseteq G_i \setminus (i+1)$, each F_i is stationary. Moreover, whenever S is stationary, $f \restriction T$ is constant for some stationary $T \subseteq S$, say with value i_0, so $T \cap F_{i_0}$ is stationary. Now let $\sigma\colon \omega_1 \times \omega_1 \to \omega_1$ be a bijection and let

$$D_i := \bigcup_{j\in\omega_1} F_{\sigma(i,j)} \text{ and } E_i := \bigcup_{j\in\omega_1} F_{\sigma(j,i)}$$

Then $(D_i)_{i\in\omega_1}$ and $(E_i)_{i\in\omega_1}$ are maximal partitions of ω_1 into stationary sets and for every $i, j \in \omega_1$, $D_i \cap E_j$ is stationary. Consequently:

Claim. *$\overline{D}$ and $\overline{E}$ are incomparable in $\leq^*$.*

Proof. Let h, C witness $\overline{D} \leq^* \overline{E}$. Let $j \neq h(0)$. Then there is $\alpha \in D_0 \cap E_j \cap C$. But then α witnesses that $D_0 \cap C \not\subseteq E_{h(0)}$. □

Ergo we can find a forcing extension in which $\mathrm{SRP}(\mathcal{S}((D_i)_{i\in\omega_1}, \omega_2, \Theta)) \wedge \neg F^+((E_i)_{i\in\omega_1}, \omega_2)$ holds and a (necessarily different) forcing extension in which $\mathrm{SRP}(\mathcal{S}((E_i)_{i\in\omega_1}, \omega_2, \Theta)) \wedge \neg F^+((D_i)_{i\in\omega_1}, \omega_2)$ holds.

6. F and F^+ at large cardinals

In this section we investigate the connections between F, F^+ and large cardinal properties, focusing in particular on the points at which F and F^+ can hold for the first time. In the case of F, this of course determines the possible patterns perfectly since $\neg F(\lambda, \kappa)$ implies $\neg F(\lambda, \kappa')$ whenever $\kappa' < \kappa$.

For simplicity, we will focus on the notationally easiest cases, i.e. $F(2, \delta)$ and $F^+((D_i)_{i\in\omega_1}, \delta)$, where $D_0 = \omega_1$, $D_i = \emptyset$ for $i \neq 0$ (it will be straightforward to see

that the proofs also work for more complicated instances of the properties). Recall that we denoted those properties by $F(\delta)$ and $F^+(\delta)$ respectively. Due to this, we also let $\mathbb{P}_F(\delta) := \mathbb{P}_{F(2)}(\delta)$ and let $\mathbb{P}_{F^+}(\delta)$ consist of functions $p\colon \gamma \to 2$ for $\gamma < \delta$ such that $p(\alpha) = 1$ implies that α has countable cofinality and p is not constant on any closed copy of ω_1.

We also fix the following definition:

Definition 6.1. Let δ be an ordinal. $f\colon \delta \to 2$ is an F-function if f is not constant on any closed copy of ω_1. $A \subseteq E^\delta_\omega$ is an F^+-set if it is stationary and does not contain a closed copy of ω_1.

We start by proving some connections between the Friedman properties and large cardinal notions in ZFC. Later on we will prove corresponding independence statements. Interestingly, the differences between F and F^+ will show up twice: On one hand, the complexity of the statement "A is stationary" leads us to weaker ZFC results while the stronger properties of the forcing $\mathbb{P}_{F^+}$ as opposed to the forcing $\mathbb{P}_F$ allows us to obtain stronger independence results, so that we always obtain a complete characterization.

Theorem 6.2. *Assume κ is weakly compact and $F(\delta)$ fails for all regular cardinals $\delta < \kappa$. Then $F(\kappa)$ fails.*

Proof. For any regular $\delta < \kappa$, let $f_\delta\colon \delta \to 2$ be a function witnessing the failure of $F(\delta)$. Let T be the tree generated by f_δ for $\delta < \kappa$. By weak compactness there is a branch $b\colon \kappa \to 2$ such that $b \restriction \alpha \in T$ for any $\alpha < \kappa$.

Claim. *b witnesses that $F(\kappa)$ fails.*

Proof. Otherwise let $c\colon \omega_1 \to \kappa$ be such that $b \restriction c[\omega_1]$ is constant. Let $\sup(c[\omega_1]) =: \alpha < \kappa$. There is a regular $\delta < \kappa$ such that $b \restriction \alpha = f_\delta \restriction \alpha$. However, then c witnesses that f_δ does not witness $\neg F(\delta)$, a contradiction. □

Ergo $F(\kappa)$ fails. □

For embedding-based large cardinal notions, we can observe that the failure persists until one cardinal above the target of the critical point using a well-known observation:

Theorem 6.3. *Assume $\kappa \leq \lambda$ are cardinals such that κ is λ-supercompact. Assume $F(\delta)$ fails for all regular cardinals $\delta < \kappa$. Then $F(\lambda^+)$ fails.*

Proof. Let $j\colon V \to M$ be a λ-supercompact embedding, i.e. ${}^\lambda M \subseteq M$ and $j(\kappa) > \lambda$. As ${}^\lambda M \subseteq M$, $(\lambda^+)^M = \lambda^+$ and since $j(\kappa)$ is inaccessible in M, $j(\kappa) > \lambda^+$. So $(F(\lambda^+))^M$ fails, witnessed by some $f\colon \lambda^+ \to 2$, $f \in M \subseteq V$. Assume $c \subseteq \lambda^+$ is closed and has ordertype ω_1 such that $f \restriction c$ is constant. Then $c \in M$ by the closure of M, a contradiction. □

For F^+, less is provable even regarding the first point of failure due to the added complexity of "A is stationary". We first prove the following ZFC result:

Theorem 6.4. *Assume $\kappa \leq \lambda$ are cardinals such that κ is λ-supercompact. Assume $F^+(\delta)$ fails for all regular cardinals $\delta < \kappa$. Then $F^+(\lambda)$ fails.*

Proof. Let $j\colon V \to M$ be a λ-supercompact embedding, i.e. ${}^\lambda M \subseteq M$ and $j(\kappa) > \lambda$. Then $(F^+(\lambda))^M$ fails, witnessed by some $A \subseteq \lambda$. In M, A is a stationary subset of E^λ_ω, which implies that A is a stationary subset of E^λ_ω in V by the closure of M (any club witnessing the nonstationarity of A in V would be in M). Ergo A witnesses that $F^+(\lambda)$ fails in V, since any closed $c \subseteq A$ with ordertype ω_1 would be in M. □

So in particular, if κ is measurable, the negation of F^+ is compact at κ.

Now we present related consistency results which show that our previous theorems for supercompact cardinals cannot be improved.

Theorem 6.5. *Assume $F^+(\delta)$ holds for all regular $\delta \geq \omega_2$ and* GCH *holds above ω_2. Let $\kappa \leq \lambda$ be cardinals such that κ is λ^+-supercompact. There is a forcing extension where κ is λ-supercompact, $F(\delta)$ fails for all regular $\delta \in [\omega_2, \lambda^+]$ and $F^+(\lambda^{++})$ holds.*

We need the following preservation result:

Lemma 6.6. *Assume δ is a regular cardinal such that $F^+(\delta)$ holds and $\mathbb{P}$ is a poset with $|\mathbb{P}| < \delta$ that does not collapse ω_1. Then $F^+(\delta)$ holds after forcing with $\mathbb{P}$.*

Proof. Assume $\dot{S}$ is a $\mathbb{P}$-name for a stationary subset of E^δ_ω. For $p \in \mathbb{P}$, let $S_p := \{\alpha \in \delta \mid p \Vdash \check{\delta} \in \dot{S}\}$. Since $\dot{S}$ is forced to be a subset of $\bigcup_{p\in\mathbb{P}} S_p$, one of the S_p is stationary. Since $F^+(\delta)$ holds, there is a closed set $c \subseteq S_p$ with ordertype ω_1. Ergo p forces that $\check{c}$ is a closed subset of $\dot{S}$ with ordertype $\dot{\omega}_1$ (since $\mathbb{P}$ does not collapse ω_1). □

Proof of Theorem 6.5. Let $(\mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha)_{\alpha\leq\lambda^+}$ be an Easton-support iteration such that $\dot{\mathbb{Q}}_\alpha$ is a $\mathbb{P}_\alpha$-name for $\mathbb{P}_F(\alpha)$ whenever α is regular and a $\mathbb{P}_\alpha$-name for the trivial poset otherwise. Let $\mathbb{P} := \mathbb{P}_{\lambda^++1} := \mathbb{P}_{\lambda^+} * \dot{\mathbb{Q}}_{\lambda^+}$.

Claim. *For any $\alpha \leq \lambda^+ + 1$, the following holds:*

(1) *$\mathbb{P}_\alpha$ forces that $F(\delta)$ fails for all regular $\delta < \alpha$.*
(2) *$\mathbb{P}_\alpha$ does not collapse any cardinals.*
(3) *$\mathbb{P}_\alpha$ forces* GCH *above ω_2.*

Proof. We do the proof by induction on α.

Let $\alpha := \beta + 1$. In case β is not regular, $\mathbb{P}_\beta$ is essentially the same as $\mathbb{P}_\alpha$, so the statement is clear (because we also have no new requirements in (1)). Assume β is regular. Then $\mathbb{P}_\alpha = \mathbb{P}_\beta * \mathbb{P}_F(\check{\beta})$. By the induction hypothesis (1), in $V[\mathbb{P}_\beta]$, $\mathbb{P}_F(\beta)$ is $< \beta$-strategically closed and has size β (by the induction hypothesis (3)). Ergo it forces $\neg F(\beta)$, does not collapse any cardinals and does not increase powerset sizes (by a nice name argument). This shows that all three statements hold for α.

Now assume α is a limit ordinal. Since $|\mathbb{P}_\alpha| \leq |\alpha|$, it suffices to show that for any cardinal $\delta < \alpha$, $\mathbb{P}_\alpha$ forces $2^\delta = \delta^+$, does not collapse δ and forces $\neg F(\delta)$ if δ is regular. So let $\delta < \alpha$. Then $\mathbb{P}_\alpha$ decomposes as $\mathbb{P}_{\delta+1} * \mathbb{P}_{\delta+1,\alpha}$, where a similar analysis as before and the usage of Easton support shows that $\mathbb{P}_{\delta+1,\alpha}$ is $\delta+1$-strategically closed in $V[\mathbb{P}_{\delta+1}]$. Ergo any subset of δ added by $\mathbb{P}_\alpha$ has been added by $\mathbb{P}_{\delta+1}$. Then the induction hypothesis for $\delta + 1$ shows that the required statements hold. □

Lastly, we show the following:

Claim. *For each $\alpha \leq \lambda^+ + 1$ that is not a singular limit cardinal, $\mathbb{P}_\alpha$ has a dense subset of cardinality $\leq |\alpha|$.*

Proof. We define the set inductively. Assume $\mathbb{R}_\beta \subseteq \mathbb{P}_\beta$ has been defined for each $\beta < \alpha$.

If $\alpha = \beta + 1$ for β nonregular, we simply let $\mathbb{R}_{\beta+1}$ consist of $(p, \emptyset)$ for $p \in \mathbb{R}_\beta$. If β is regular, we let $\mathbb{R}_{\beta+1}$ consist of (p, τ), where $p \in \mathbb{R}_\beta$ and τ is a nice $\mathbb{R}_\beta$-name for an element of $\dot{\mathbb{Q}}_\beta$. It follows from GCH above ω_2, the size of $\dot{\mathbb{Q}}_\beta$ being forced to equal β and the fact that $|\mathbb{R}_\beta| = \beta$ that $|\mathbb{R}_{\beta+1}| = \beta$ and it is clearly dense.

If α is a limit, we let $\mathbb{R}_\alpha$ consist of those $f \in \mathbb{P}_\alpha$ such that $f \restriction \beta \in \mathbb{R}_\beta$ for each $\beta < \alpha$. We see as follows that $\mathbb{R}_\alpha$ is dense: Given any $p \in \mathbb{P}_\alpha$, let q be defined as follows: If $q \restriction \beta$ has been constructed, let $q(\beta)$ be such that $(q \restriction \beta, q(\beta)) \in \mathbb{R}_{\beta+1}$

and $q \restriction \beta \Vdash q(\beta) = p(\beta)$ (this is possible by the definition of $\mathbb{R}_{\beta+1}$). This also shows that in limit steps, $q \restriction \beta$ will be in $\mathbb{R}_\beta$, so $q \in \mathbb{R}_\alpha$.

Clearly, if α is not a limit cardinal, $|\mathbb{R}_\alpha| = |\mathbb{R}_{\beta+1}|$ where β is the largest cardinal below α, so $|\mathbb{R}_\alpha| = \beta = |\alpha|$. If α is a regular limit cardinal, we can identify $\mathbb{R}_\alpha$ with the set of all $<\alpha$-sized partial functions from α to α, so $|\mathbb{R}_\alpha| = \alpha$ by GCH above ω_2. □

For simplicity, assume $\mathbb{P}$ is equal to its λ^+-sized dense subset. So $F^+(\lambda^{++})$ holds after forcing with $\mathbb{P}$ by Lemma 6.6. The only thing left is to show that $\mathbb{P}$ forces that κ is λ-supercompact. To this end, let G be $\mathbb{P}$-generic and let $j: V \to M$ be a λ^+-supercompact embedding, i.e. ${}^{\lambda^+}M \subseteq M$ and $j(\kappa) > \lambda^+$. Write $G = G_\kappa * G_{\kappa,\lambda^+}$. We note that $j(\mathbb{P}) \cong \mathbb{P} * j(\mathbb{P})_{\lambda^+ +1, j(\lambda^+ +1)}$, since $j(\mathbb{P})_{\lambda^+ +1} = \mathbb{P}$ by the closure of M. Let $H_{j(\kappa)}$ be any $j(\mathbb{P})_{j(\kappa)}$-generic filter containing G. Since $j[G_\kappa] = G_\kappa \subseteq H_{j(\kappa)}$, j lifts to $j^+: V[G_\kappa] \to M[H_{j(\kappa)}]$.

By the closure of M, M contains $j \restriction \mathbb{P}_{\kappa+1,\lambda^+ +1}$ and therefore $M[H_{j(\kappa)}]$ contains $j[G_{\kappa,\lambda^+}]$.

Claim. *In $M[H_{j(\kappa)}]$, there exists $q \in j(\mathbb{P})_{j(\kappa)+1, j(\lambda^+)+1}$ which is a lower bound of $j[G_{\kappa,\lambda^+}]$.*

Proof. We will show that $j[G_{\kappa,\lambda^+}]$ is $<\omega_2$-directed in $M[H_{j(\kappa)}]$. Since $|j[G_{\kappa,\lambda^+}]| = \lambda^+ < j(\kappa)$ and $j(\mathbb{P})_{j(\kappa)+1, j(\lambda^+)+1}$ is $<j(\kappa)$-$<\omega_2$-directed closed by Lemma 2.13 and Theorem 2.15, this suffices.

So let $A \subseteq j[G_{\kappa,\lambda^+}]$ have size ω_1. Then $j^{-1}[A] \in V[H_{j(\kappa)}]$ has size ω_1 as well and is therefore in $V[G_\kappa] \subseteq V[G]$. In $V[G]$, G_{κ,λ^+} is $<\omega_2$-directed by Lemma 2.14 and therefore there is $q \in G_{\kappa,\lambda^+}$ which is a lower bound of $j^{-1}[A]$. By elementarity, $j(q) \in j[G_{\kappa,\lambda^+}]$ is a lower bound of $j(j^{-1}[A]) = j[j^{-1}[A]] = A$. □

Now let $H_{j(\kappa),j(\lambda^+)}$ be any $j(\mathbb{P})_{j(\kappa)+1, j(\lambda^+)+1}$-generic filter containing q and let $H := H_{j(\kappa)} * H_{j(\kappa),j(\lambda^+)}$. Then $j[G] \subseteq H$ and so j lifts to $j^+: V[G] \to M[H]$ in $V[H]$. In particular, in $V[H]$, there exists a κ-complete normal and fine ultrafilter over $(\mathcal{P}([\lambda]^{<\kappa}))^{V[G]}$. This ultrafilter has size $2^{\lambda^{<\kappa}} = \lambda^+$ by assumption. Since $V[H]$ is an extension of $V[G]$ using a $<\lambda^{++}$-strategically closed forcing (once again, by the closure of M), this ultrafilter is in $V[G]$. Ergo, κ is λ-supercompact in $V[G]$. □

We can also show that the propagation of F^+ at supercompact cardinals is optimal:

Theorem 6.7. *Assume $\kappa \leq \lambda$ are regular cardinals such that κ is λ-supercompact. Assume* GCH *holds above ω_2 as well as $F^+(\delta)$ for all regular $\delta \geq \omega_2$. There is a forcing extension where κ remains λ-supercompact, $F^+(\delta)$ fails for all regular $\delta \leq \lambda$ and $F^+(\lambda^+)$ holds.*

Proof. Let $\mathbb{P} := (\mathbb{P}_\alpha, \dot{\mathbb{Q}}_\alpha)_{\alpha<\lambda+1}$ be an Easton support iteration where $\dot{\mathbb{Q}}_\alpha$ is a $\mathbb{P}_\alpha$-name for $\mathbb{P}_{F^+}(\alpha)$ if $\alpha \geq \omega_2$ is regular and for the trivial poset otherwise. Then, just like before, it follows that $\mathbb{P}$ preserves all cofinalities and powerset sizes and forces that $F^+(\delta)$ fails for all regular $\delta \leq \lambda$ (since the stationarity of any added set is preserved thanks to the closure of the tail forcing). Furthermore, as in the proof of Theorem 6.5 it follows that $\mathbb{P}$ forces $F^+(\lambda^+)$ to hold. The only thing left to show is that κ remains λ-supercompact in $V[G]$, where G is $\mathbb{P}$-generic. Write $G = G_\kappa * G_{\kappa,\lambda}$ and let $j: V \to M$ be a λ-supercompact embedding. We first construct a master condition as before and then use the stronger properties of $\mathbb{P}_{F^+}$ to build a sufficiently generic filter in $V[G]$ which allows us to lift the embedding.

First note that $j(\mathbb{P}) \cong \mathbb{P} * j(\mathbb{P})_{\lambda+1, j(\lambda)+1}$. Let $H_{j(\kappa)}$ be any $j(\mathbb{P})_{j(\kappa)}$-generic filter containing G. Then $j[G] = G \subseteq H_{j(\kappa)}$ and so j lifts to $j^+: V[G_\kappa] \to M[H_{j(\kappa)}]$. As before, $j[G_{\kappa,\lambda}] \in M[H_{j(\kappa)}]$.

We have a similar claim to before:

Claim. *In $M[H_{j(\kappa)}]$, there is a condition $q \in j(\mathbb{P})_{\lambda+1,j(\lambda+1)}$ which is a lower bound of $j[G_{\kappa,\lambda}]$.*

Proof. As before, we will show that $j[G_{\kappa,\lambda}]$ is $<\omega_2$-directed in $M[H_{j(\kappa)}]$. Since $|j[G_{\kappa,\lambda}]|= \lambda < j(\kappa)$ and $j(\mathbb{P})_{j(\kappa)+1,j(\lambda)+1}$ is $<j(\kappa)$-$<\omega_2$-directed closed by Lemma 2.13 and Theorem 2.15, this suffices.

So let $A \subseteq j[G_{\kappa,\lambda}]$ have size $\leq \omega_1$. Then $j^{-1}[A] \in V[H_{j(\kappa)}]$ has size $\leq \omega_1$ as well and is therefore in $V[G_\kappa] \subseteq V[G]$. In $V[G]$, $G_{\kappa,\lambda}$ is $<\omega_2$-directed by Lemma 2.14 and therefore there is a lower bound $q \in G_{\kappa,\lambda}$. By elementarity, $j(q) \in j[G_{\kappa,\lambda}]$ is a lower bound of $j(j^{-1}[A]) = j[j^{-1}[A]] = A$. □

$\mathbb{P}$ contains a dense set $\mathbb{R}$ of cardinality λ and thus has the λ^+-cc.. Ergo, there are at most λ^+ many maximal antichains of $\mathbb{P}$ contained in $\mathbb{R}$. By elementarity, there are at most $j(\lambda^+)$ many maximal antichains of $j(\mathbb{P})$ contained in $j(\mathbb{R})$ lying in M. However, $j(\lambda^+)$ is equal to the set of all equivalence classes of functions from $\lambda^{<\kappa}$ to λ, so $|j(\lambda^+)|= 2^\lambda = \lambda^+$ in V (which is of course not changed by forcing extensions). Ergo in $V[G]$ we can find an enumeration $(A_\alpha)_{\alpha<\lambda^+}$ of all maximal antichains of $j(\mathbb{P}) \restriction [\lambda+1, j(\lambda+1))$ lying in $M[G]$. Using the fact that $j(\mathbb{P}) \restriction [\lambda+1, j(\lambda+1))$ is λ^+-strategically closed we can now find a descending sequence $(q_\alpha)_{\alpha<\lambda^+}$ in $V[G]$ of elements of $j(\mathbb{P}) \restriction [\lambda+1, j(\lambda+1))$ such that $q_0 \leq q$ and for any $\alpha < \lambda^+$, q_α is below an element of A_α. Ergo the upward closure of $\{q_\alpha \mid \alpha < \lambda^+\}$, denote it by H, is a filter for $j(\mathbb{P}) \restriction [\lambda+1, j(\lambda+1))$ that is generic over $M[G]$ (by density of $j(\mathbb{R})$). Furthermore, $H \in V[G]$. Ergo, in $V[G]$ we can lift the embedding j to $j^+\colon V[G] \to M[G*H]$ which shows that κ is λ-supercompact in $V[G]$. □

Remark 6.8. As before we can see that, while λ^+-strategic closure (and even strong $<\lambda^+$-distributivity) is sufficient to build suitable generic filters, the same is not true for $<\lambda^+$-strategic closure.

7. Open Questions

We close with a few open questions.

An earlier version of the manuscript contained an erroneous proof of the non-compactness of F at Mahlo cardinals and of F^+ at weakly compact cardinals. The issues were (1) the existence of ghost coordinates in trying to replicate the proof of the consistency of $\mathrm{MM}_{F(\lambda)}(f)$ due to the fact that any iteration forcing $\neg F(\delta)$ for $\delta < \kappa$ Mahlo necessarily contains conditions with support $\geq \omega_2$ and (2) the non-existence of an analogue of Lemma 6.6 for forcings which were merely δ-cc instead of having size $<\delta$. Thus we leave the following questions open, noting that we expect them to have a negative answer.

Question 7.1. Suppose κ is Mahlo and $F(\delta)$ fails for all $\delta < \kappa$. Must $F(\kappa)$ fail?

Question 7.2. Suppose κ is weakly compact and $F^+(\delta)$ fails for all $\delta < \kappa$. Must $F^+(\kappa)$ fail?

The following would simplify our proof of the consistency of $F^+(\omega_2)$ and is of independent interest:

Question 7.3. Suppose κ is Mahlo. Is there a forcing extension in which κ remains Mahlo and there is a sequence $(A_\alpha)_{\alpha<\kappa}$ such that whenever $S \subseteq \kappa$ is a stationary set of singular ordinals and $A \subseteq \kappa$, there is $\nu < \kappa$ regular such that $S \cap \nu$ is stationary in ν and $A \cap \nu = A_\nu$?

If κ is 1-Mahlo (i.e. κ is regular and the set of Mahlo cardinals is stationary in κ), there is a forcing extension in which κ is 1-Mahlo and there is a sequence $(A_\alpha)_{\alpha<\kappa}$

such that whenever $S \subseteq \kappa$ is a stationary set of singular ordinals and $A \subseteq \kappa$, there is $\nu < \kappa$ Mahlo such that $S \cap \nu$ is stationary in ν and $A \cap \nu = A_\nu$. This model is obtained by first forcing that every stationary subset of $\mathrm{Sing} \cap \kappa$ reflects to a Mahlo cardinal and then adding a diamond sequence via the Easton-support version of Cohen forcing at κ (i.e. conditions are partial functions $p\colon \kappa \to 2$ such that $\mathrm{dom}(p)$ is an Easton set, ordered by extension). This poset is nicely κ-cc if κ is 1-Mahlo and so Lemma 1.12 applies. If κ is merely Mahlo, this does not hold.

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